

VERTEX ALGEBRAS

github.com/danimalabares/vertex-algebras

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1. CARTAN SUBALGEBRA, CARTAN MATRIX AND SERRE RELATIONS

Kac-Moody algebras are Lie algebras, whose definition is motivated by the structure of finite-dimensional simple Lie algebras over $\mathbb{C}$.

Let $\mathfrak{g}$ be a finite-dimensional semisimple Lie algebra over $\mathbb{C}$. Then $\mathfrak{g}$ has a *Cartan subalgebra* $\mathfrak{h} \subset \mathfrak{g}$ (abelian + ...). Fixing $\mathfrak{h} \subset \mathfrak{g}$ gives a *root space decomposition*

$$\mathfrak{g} = \mathfrak{h} \oplus \bigoplus_{\alpha \in \Delta} \mathfrak{g}_{\alpha}$$

where $\Delta \subset \mathfrak{h}^*$ linear dual, and, by definition

$$\mathfrak{g}_{\alpha} = \{X \in \mathfrak{g} \mid [H, X] = \alpha(H)X \ \forall H \in \mathfrak{h}\}$$

Turns out the $\mathfrak{g}_{\alpha}$ are all 1-dimensional, though this property is lost when we go to Kac-Moody algebras.

$$[\mathfrak{g}_{\alpha}, \mathfrak{g}_{\beta}] \subset \mathfrak{g}_{\alpha+\beta}$$

The Killing form $\kappa : \mathfrak{g} \times \mathfrak{g} \rightarrow \mathbb{C}$, $\kappa(x, y) = \text{Tr}_{\mathfrak{g}} \text{ad}(x)\text{ad}(y)$ is nondegenerate. “This is kind of the definition of semisimple.” (Think of $\mathfrak{h}$ as $\mathfrak{g}_0$, btw.)

$\kappa|_{\mathfrak{g}_{\alpha} \times \mathfrak{g}_{\beta}} \neq 0$ only when $\beta = -\alpha$. $\kappa|_{\mathfrak{h} \times \mathfrak{h}}$ is non-degenerate. This gives a linear isomorphism $\mathfrak{h} \xrightarrow{\nu} \mathfrak{h}$ via $\nu(H)(H') = \kappa(H, H')$.

So, $\mathfrak{h}^*$ comes with a non-degenerate bilinear form.

The *reflection* $r_{\alpha} : \mathfrak{h}^* \rightarrow \mathfrak{h}^*$ in $\alpha \in \mathfrak{h}^*$ (usually a root) is $r_{\alpha}(\lambda) = \lambda - 2 \frac{(\lambda, \alpha)}{(\alpha, \alpha)} \cdot \alpha$.

“Classify root systems [...] classify semisimple Lie algebras” It is a fact that $r_\alpha(\Delta) = \Delta$ for all $\alpha \in \Delta$, which motivates the definition of *root system* and permits classification.

Example 1.1. $\mathfrak{g} = \mathfrak{sl}_2$, $\mathfrak{h} =$ diagonal matrices

$$\begin{pmatrix} 1 & & \\ & -1 & \\ & & 0 \end{pmatrix} \quad \begin{pmatrix} 0 & & \\ & 1 & \\ & & -1 \end{pmatrix}$$

is a basis of $\mathfrak{h}$. There are 6 roots vectors

$$E_{12} = \begin{pmatrix} 0 & 1 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix}$$

E_{23}, E_{13} , etc.

Exercise 1.2. $[H_1, E_{12}] = 2E_{12}$, $[H_2, E_{12}] = -E_{12}$, $\alpha_{12} = (2, -1)$.

[Drawing of roots]

Notions of *positive roots* and *simple roots* (set of rank $\mathfrak{g}$ simple roots has ℓ elements, where $\ell = \dim(\mathfrak{h}^*)$). This will also fail for Kac-Moody algebras more generally). Next write the *Cartan matrix*

$$A = (a_{ij}), \quad a_{ij} = 2 \frac{(\alpha_i, \alpha_j)}{(\alpha_i, \alpha_i)}$$

for $1 \leq i, j \leq \ell$.

Example 1.3. $\mathfrak{sl}_3$. [Picture, hexagonal pattern]. $(\alpha_1, \alpha_1) = (\alpha_2, \alpha_2) = 2$, $(\alpha_1, \alpha_2) = -1$, so

$$A = \begin{pmatrix} 2 & -1 \\ -1 & 2 \end{pmatrix}$$

Example 1.4. $\mathfrak{sl}_5$. [Picture, square pattern]. $|\alpha_2| = 1$, $|\alpha_1| = 2$, $(\alpha_1, \alpha_2) = -2$, so

$$A = \begin{pmatrix} 2 & -1 \\ -2 & 2 \end{pmatrix}$$

Remark 1.5. Since $\mathfrak{g}_\alpha$ is 1-dimensional, set $\mathfrak{g}_\alpha = \mathbb{C}E_\alpha$ and $E_i = E_{\alpha_i}$, $i = 1, 2, \dots, \ell$ (simple root vectors). It turns out that

$$\text{ad}(E_i)^{1-a_{ij}} E_j = 0.$$

This is called a *Serre relation*.

2. SOME INFINITE DIMENSIONAL LIE ALGEBRAS

Let $\mathfrak{g}$ be a finite-dimensional semisimple Lie algebra, and define the *loop algebra*

$$\begin{aligned} L\mathfrak{g} &= \mathfrak{g}[t, t^{-1}], \text{ (with basis } at^m | a \in \text{a basis of } \mathfrak{g} \text{)} \\ &= \mathfrak{g} \otimes_{\mathbb{C}} \mathbb{C}[t, t^{-1}] \end{aligned}$$

with the Lie bracket

$$[at^m, bt^n] = [a, b]t^{m+n}.$$

“This construction is absurdly general — we don’t need $\mathfrak{g}$ to be semisimple [...]”

Take $\mathfrak{g} = \mathfrak{sl}_2$. Recall that

$$E = \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}, \quad H = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}, \quad F = \begin{pmatrix} 0 & 0 \\ 1 & 0 \end{pmatrix}$$

[Picture with $F, H, E, Ft, Ht, Et, Et^2 \dots$] E was a root vector, corresponding to the unique root in $\mathfrak{sl}_2$, call it α_1 . We seem to have a second simple root α_0 , corresponding to Ft .

This looks like it wants to have a Cartan matrix

$$A = \begin{pmatrix} 2 & -2 \\ -2 & 2 \end{pmatrix}$$

We will indeed recover (a variant of) $L\mathfrak{g}$ as a Lie algebra “built from” $A = \begin{pmatrix} 2 & -2 \\ -2 & 2 \end{pmatrix}$, a Kac-Moody algebra. But note first, $\mathfrak{h} = \mathbb{C}H$ is too small. “Problem with α_0 and α_1 being linearly independent ...”

Exercise 2.1. Consider $L\mathfrak{g} \oplus \mathbb{C}d$, and set $[d, at^m] = mat^m$, $[d, d] = 0$. Check this defines a Lie algebra.

Proof. Skew-commutativity, i.e. for all $x \in L\mathfrak{g} \oplus \mathbb{C}d$,

$$(2.1.1) \quad [x, x] = 0,$$

is immediate from skew commutativity in $L\mathfrak{g}$ and the hypothesis that $[d, d] = 0$.

To confirm Jacobi identity, i.e. that for all $x, y, z \in L\mathfrak{g} \oplus \mathbb{C}d$

$$(2.1.2) \quad [x, [y, z]] + [y, [z, x]] + [z, [x, y]] = 0,$$

notice that since this is a cyclic sum on x, y, z we only need to consider three elements in $L\mathfrak{g} \oplus \mathbb{C}d$ up to cyclic permutation. The cases in which the three elements are either in $L\mathfrak{g}$ or in $\mathbb{C}d$ are obvious, so that there are only two interesting possibilities:

$$(2.1.3) \quad x = d, \quad y = at^m, \quad z = bt^n$$

$$(2.1.4) \quad x = d, \quad y = d, \quad z = at^n$$

Case 2.1.3 gives

$$\begin{aligned} & [d, [at^m, bt^n]] + [at^m, [bt^n, d]] + [bt^n, [d, at^m]] \\ &= [d, [a, b]t^{m+n}] + [at^m, -nbt^n] + [bt^n, mat^m] \\ &= (m+n)[a, b]t^{m+n} - n[a, b]t^{m+n} + m[b, a]t^{m+n} \\ &= (m+n)[a, b]t^{m+n} - n[a, b]t^{m+n} - m[a, b]t^{m+n} \\ &= (m+n)[a, b]t^{m+n} - (m+n)[a, b]t^{m+n} = 0. \end{aligned}$$

Case 2.1.4 gives

$$\begin{aligned} & [d, [d, at^m]] + [d, [at^m, d]] + [at^m, [d, d]] \\ &= [d, mat^m] + [d, -mat^m] = 0. \end{aligned}$$

□

The Kac-Moody algebra turns out to be, not quite this, but slightly larger still.

Recall that an *invariant bilinear form* $(\cdot, \cdot)$ on a Lie algebra $\mathfrak{g}$ is a bilinear form such that

$$(2.1.5) \quad ([a, b], c) = (a, [b, c]) \quad \forall a, b, c \in \mathfrak{g}.$$

Exercise 2.2. Prove that an invariant bilinear form on a simple Lie algebra must in fact be symmetric.

Proof. It's enough to show that $\mathfrak{g}$ is *perfect*, i.e. that $[\mathfrak{g}, \mathfrak{g}] = \mathfrak{g}$. In this case, let $a, b \in \mathfrak{g}$ and suppose that $b = [x, y]$. Then

$$\begin{aligned} (a, b) &= (a, [x, y]) = (a, -[y, x]) = (-[a, y], x) = ([y, a], x) \\ &= (y, [a, x]) = (y, -[x, a]) = (-[y, x], a) = ([x, y], a) = (b, a) \end{aligned}$$

To confirm that $\mathfrak{g}$ is perfect just observe that $[\mathfrak{g}, \mathfrak{g}]$ is a nontrivial ideal of $\mathfrak{g}$. $\square$

Definition 2.3. Given $\mathfrak{g}$ simple, with $(\cdot, \cdot) : \mathfrak{g} \times \mathfrak{g} \rightarrow \mathbb{C}$ invariant bilinear form, the *affine Lie algebra* is

$$\hat{\mathfrak{g}} = L\mathfrak{g} \oplus \mathbb{C}K,$$

with $[K, \hat{\mathfrak{g}}] = 0$, and $[at^m, bt^n] = [a, b]t^{m+n} + m(a, b)\delta_{m, -n}K$.

“For the construction to work it doesn't actually have to be nondegenerate.”

Exercise 2.4. Check that the affine Lie algebra $\hat{\mathfrak{g}}$ is a Lie algebra.

Proof. (Skew-commutativity.) Since $[K, \hat{\mathfrak{g}}] = 0$ and $K \in \hat{\mathfrak{g}}$, it is immediate that $[K, K] = 0$. For the case of an element in $L\mathfrak{g}$, we see that $[at^m, at^m] = 0$ by skew-commutativity of the bracket in $\mathfrak{g}$ and the Kronecker delta.

(Jacobi identity.) As in Exercise 2.1, any choice of x, y, z involving K is immediate by $[K, \hat{\mathfrak{g}}] = 0$. Thus the only interesting case is for Jacobi identity consider the cases

$$\begin{aligned} &[at^m, [bt^n, ct^\ell]] + [bt^n, [ct^\ell, at^m]] + [ct^\ell, [at^m, bt^n]] \\ &= [at^m, [b, c]t^{n+\ell} + n(b, c)\delta_{n, -\ell}K] \\ &+ [bt^n, [c, a]t^{\ell+m} + \ell(c, a)\delta_{\ell, -m}K] \\ &+ [ct^\ell, [a, b]t^{m+n} + m(a, b)\delta_{m, -n}K] \\ &= [at^m, [b, c]t^{n+\ell}] + [at^m, n(b, c)\delta_{n, -\ell}K] \\ &+ [bt^n, [c, a]t^{\ell+m}] + [bt^n, \ell(c, a)\delta_{\ell, -m}K] \\ &+ [ct^\ell, [a, b]t^{m+n}] + [ct^\ell, m(a, b)\delta_{m, -n}K] \\ &= [a, [b, c]]t^{m+(n+\ell)} + m(a, [b, c])\delta_{m, -(n+\ell)}K \\ &+ [b, [c, a]]t^{n+(\ell+m)} + n(b, [c, a])\delta_{n, -(\ell+m)}K \\ &+ [c, [a, b]]t^{\ell+(m+n)} + \ell(c, [a, b])\delta_{\ell, -(m+n)}K = 0 \end{aligned}$$

It is clear that we obtain a Jacobi equation on $\mathfrak{g}$. To see that the remaining terms vanish, notice that the condition on the Kronecker delta in its three appearances is the same, namely, $m + n + \ell = 0$. In this case, we only need to check that $(a, [b, c]) = (b, [c, a]) = (c, [a, b])$ to conclude. This follows from the invariance of $(\cdot, \cdot)$ and the fact that $\mathfrak{g}$ simple using Exercise 2.2. $\square$

We also have

Definition 2.5. The *extended affine Lie algebra* is

$$\tilde{\mathfrak{g}} = L\mathfrak{g} \oplus \mathbb{C}K \oplus \mathbb{C}d,$$

with $[d, at^m] = mat^m$ as before, and $[K, d] = 0$.

The extended affine Lie algebra is an example of a Kac-Moody algebra.

Exercise 2.6 (For those who like geometry). Let $R = \mathbb{C}[t, t^{-1}]$. If $D \in \text{Der}(R)$, then $L\mathfrak{g} \oplus \mathbb{C}d$ is a Lie algebra with $[d, a \otimes r] = a \otimes D(r)$. Is $(\mathfrak{g} \otimes R) \oplus \text{Der}(R)$ a Lie algebra? (The Lie algebra $L\mathfrak{g} \oplus \mathbb{C}d$ from Exercise 2.1 is a particular case, for $D = t \frac{d}{dt}$.)

Proof. Checking that $L\mathfrak{g} \oplus \mathbb{C}d$ is a Lie algebra with $[d, a \otimes r] = a \otimes D(r)$ is similar to Exercise 2.1: skew-commutativity is immediate from skew-commutativity in each of the components, while Jacobi identity is verified in two cases. For $x = y = d$ and $z = a \otimes r$ we quickly obtain

$$\begin{aligned} & [x, [y, z]] + [y, [z, x]] + [z, [x, y]] \\ &= [d, [d, a \otimes r]] + [d, [a \otimes r, d]] + [a \otimes r, [d, d]] \\ &= [d, a \otimes D(r)] + [d, -a \otimes D(r)] = 0. \end{aligned}$$

And for $x = d$, $y = a \otimes r$ and $z = b \otimes s$, we get

$$\begin{aligned} & [x, [y, z]] + [y, [z, x]] + [z, [x, y]] \\ (2.6.1) \quad &= [d, [a \otimes r, b \otimes s]] + [a \otimes r, [b \otimes s, d]] + [b \otimes s, [d, a \otimes r]] \\ &= [d, [a, b] \otimes rs] + [a \otimes r, -b \otimes D(s)] + [b \otimes s, a \otimes D(r)] \\ &= [a, b] \otimes D(rs) - [a, b] \otimes rD(s) + [b, a] \otimes sD(r) = 0. \end{aligned}$$

To check whether $(\mathfrak{g} \otimes R) \oplus \text{Der}(R)$ is a Lie algebra first put the Lie bracket on $\text{Der}(R)$ as $[D, D_1] = DD_1 - D_1D$. It is clear that this bracket is skew-commutative. Jacobi identity reads

$$\begin{aligned} & [D, [D_1, D_2]] + [D_1, [D_2, D]] + [D_2, [D, D_1]] \\ &= [D, D_1D_2 - D_2D_1] + [D_1, D_2D - DD_2] + [D_2, DD_1 - D_1D] \\ &= D(D_1D_2 - D_2D_1) - (D_1D_2 - D_2D_1)D + D_1(D_2D - DD_2) \\ &\quad - (D_2D - DD_2)D_1 + D_2(DD_1 - D_1D) - (DD_1 - D_1D)D_2 \\ &= DD_1D_2 - DD_2D_1 - D_1D_2D + D_2D_1D + D_1D_2D - D_1DD_2 \\ &\quad - D_2DD_1 + DD_2D_1 + D_2DD_1 - D_2D_1D - DD_1D_2 + D_1DD_2 = 0. \end{aligned}$$

Now put the bracket on $(\mathfrak{g} \otimes R) \oplus \text{Der}(R)$ as $[D, a \otimes r] = a \otimes D(r)$. Skew-commutativity is immediate. Jacobi identity for $x = D, y = a \otimes r$ and $z = b \otimes s$ is identical to the computation 2.6.1. In the case $x = D, y = D_1$ and $z = a \otimes r$, we get

$$\begin{aligned} & [D, [D_1, a \otimes r]] + [D_1, [a \otimes r, D]] + [a \otimes r, [D, D_1]] \\ &= [D, a \otimes D_1(r)] + [D_1, -a \otimes D(r)] + [a \otimes r, [D, D_1]] \\ &= a \otimes DD_1(r) - a \otimes D_1D(r) - a \otimes [D, D_1](r) = 0 \end{aligned}$$

□

3. KAC-MOODY ALGEBRAS

Recall the notion of the free Lie algebra on a vector space V of generators (or a set X , think of V as a vector space with basis X):

Definition 3.1. The *free Lie algebra* on V is characterized by the universal property

$$\begin{array}{ccc} V & \xrightarrow{f} & \mathfrak{g} \\ & \searrow i & \nearrow \exists! \tilde{f} \\ & F(V) & \end{array}$$

That is, for any linear map $f : V \rightarrow \mathfrak{g}$ with $\mathfrak{g}$ Lie algebra, there exists a unique $\tilde{f}$ homomorphism of Lie algebras $F(V) \rightarrow \mathfrak{g}$ such that $\tilde{f} \circ i = f$.

$$\mathrm{Hom}_{\mathrm{Lie}}(F(V), \mathfrak{g}) = \mathrm{Hom}_{\mathrm{Vec}}(V, \mathfrak{g})$$

naturally.

That is, F and the forgetful functor $G : \underline{\mathrm{Lie}} \rightarrow \underline{\mathrm{Vec}}$ are adjoint:

$$\mathrm{Hom}_{\underline{\mathrm{Lie}}}(F(V), \mathfrak{g}) \xrightarrow{\cong} \mathrm{Hom}_{\underline{\mathrm{Vec}}}(V, G(\mathfrak{g}))$$

A realisation of $F(V)$. Let

$$T(V) = \mathbb{C} \oplus V \oplus V^{\otimes 2} \oplus V^{\otimes 3} \oplus \dots$$

be the tensor algebra of V .

Then inside $T(V)$ consider $F(V)$ the span of iterated commutators of elements of V .

Proposition 3.2. *This realises the free Lie algebra.*

Proof. In online notes. □

In the finite dimensional simple case, we had

$$a_{ij} = \frac{2(\alpha_i, \alpha_j)}{(\alpha_i, \alpha_i)},$$

which we think also as $\alpha_i, \alpha_j \in \mathfrak{h}^*$, and $\alpha_i^\vee = \frac{2}{(\alpha_i, \alpha_i)} \nu^{-1}(\alpha_i) \in \mathfrak{h}$.

Clearly, $\alpha_{ii} = 2$ for all i . a_{ij} might not equal a_{ji} , but certainly $a_{ij} = 0 \iff a_{ji} = 0$. And $\forall i \neq j$, $a_{ij} \leq 0$.

One further property. Set

$$\varepsilon_i = \frac{2}{(\alpha_i, \alpha_i)}, \quad \text{and} \quad D = \begin{array}{c} \text{diagonal matrix} \\ \text{with entries } \varepsilon_i \end{array}$$

Then $A = DB$, where $B = ((\alpha_i, \alpha_j))$ is symmetric. If a matrix A is equal to $(\mathrm{diag})(\mathrm{symm})$, we call it *symmetrizable*.

Definition 3.3. A *generalized Cartan matrix* is an integer matrix $A = (a_{ij})$ which is

- symmetrizable,
- $a_{ii} = 2$ for all i ,
- $a_{ij} = 0 \iff a_{ji} = 0$,
- $a_{ij} \leq 0$ for $i \neq j$.

Definition 3.4. A *realisation* of a generalized Cartan matrix is a complex vector space $\mathfrak{h}$, and two sets

$$\begin{aligned}\Pi^\vee &= \{\alpha_1^\vee, \alpha_2^\vee, \dots, \alpha_n^\vee\}, \quad \text{and,} \\ \Pi &= \{\alpha_1, \alpha_2, \dots, \alpha_n\}\end{aligned}$$

such that $\langle \alpha_i^\vee, \alpha_j \rangle = a_{ij}$, $1 \leq i, j \leq n$.

Exercise 3.5. $\dim(\mathfrak{h}) \geq 2n - \text{rank}(A)$.

Proof. For $A = \begin{pmatrix} 2 & -2 \\ -2 & 2 \end{pmatrix}$, a realisation is given by

$$\Pi^\vee = \{H_1, H_0\}, \quad \Pi = \{\alpha_0, \alpha_1\}$$

$$\mathfrak{h} = \mathbb{C}H, \mathbb{C}d, \mathbb{C}K,$$

$$\mathfrak{h}^* = \mathbb{C}\alpha_1 + \mathbb{C}\delta + \mathbb{C}\Lambda_0$$

(Canonical dual, $\langle \alpha_1, H \rangle = 2$, $\langle \delta, d \rangle = 1 = \langle \Lambda_0, K \rangle$, every other pairing 0.)

Then

$$\begin{cases} \alpha_1 = \alpha_1 \\ \alpha_0 = \delta - \alpha_1 \end{cases} \quad \begin{cases} \alpha_1^\vee = H \\ \alpha_0^\vee = K - H \end{cases}$$

So we obtain

$$\begin{aligned}\langle \alpha_0^\vee, \alpha_1 \rangle &= \langle K - H, \alpha_1 \rangle = 2 \\ \langle \alpha_1^\vee, \alpha_0 \rangle &= \langle H, \delta - \alpha_1 \rangle = -2 \\ \langle \alpha_0^\vee, \alpha_0 \rangle &= \langle K - H, \delta - \alpha_1 \rangle = +2\end{aligned}$$

□

Finally let's define Kac-Moody algebras.

Let A be a generalized Cartan matrix. Let

$$\tilde{\mathfrak{n}}_+ = F(e_1, \dots, e_n),$$

the free Lie algebra on n generators, and similarly

$$\tilde{\mathfrak{n}}_- = F(f_1, \dots, f_n).$$

Let $\mathfrak{h}$ be a realisation of A . Set $\tilde{\mathfrak{g}}(A) = \tilde{\mathfrak{n}}_- \oplus \mathfrak{h} \oplus \tilde{\mathfrak{n}}_+$.

Make $\tilde{\mathfrak{g}}(A)$ a Lie algebra by defining

- $[\mathfrak{h}, \mathfrak{h}] = 0$,
- $\forall H \in \mathfrak{h}, [H, e_i] = \langle \alpha_i, H \rangle e_i = \alpha_i(H)e_i$. And similarly, $[H, f_i] = -\alpha_i(H)f_i$.
- $[e_i, f_j] = \delta_{ij}\alpha_i^\vee$.

Then $\tilde{\mathfrak{g}}(A)$ is a Lie algebra (though not yet the Kac-Moody algebra). See Kac, [Kac90, Theorem 1.2].

Remark 3.6. In $\mathfrak{h}$ we have a lattice

$$Q^\vee = \mathbb{Z}\alpha_1^\vee + \dots + \mathbb{Z}\alpha_n^\vee, \quad \text{and}$$

$$Q = \mathbb{Z}\alpha_1 + \dots + \mathbb{Z}\alpha_n \text{ in } \mathfrak{h}^*$$

(root and coroot lattices). $\tilde{\mathfrak{g}}(A)$ is naturally Q -graded, with

$$\tilde{\mathfrak{g}}(A)_\beta = \text{span}\{\text{commutators of } e_i \text{ with } \sum \alpha_i = \beta\}.$$

$$\tilde{\mathfrak{g}}(A) = \mathfrak{h}.$$

Theorem 3.7 (Gabber-Kac). *Denote by $I \subset \tilde{\mathfrak{g}}(A)$ the maximal Q -graded ideal, such that $I \cap \mathfrak{h} = \{0\}$. Then I is generated by the Serre relations*

$$\text{ad}(e_i)^{1-a_{ij}} e_j \quad \text{and} \quad \text{ad}(f_i)^{1-a_{ij}} f_j, \quad i \neq j.$$

Proof. [Kac90, Theorem 9.11]. $\square$

(The existence of the ideal I does not need the theorem; the importance of the theorem is providing an expression for the generators.)

Definition 3.8. The Kac-Moody algebra $\mathfrak{g}(A)$ is $\tilde{\mathfrak{g}}(A)/I$.

4. AFFINE KAC-MOODY ALGEBRAS

Let $\mathfrak{g}$ be a finite-dimensional simple Lie algebra, with $(\cdot, \cdot) : \mathfrak{g} \times \mathfrak{g} \rightarrow \mathbb{C}$ invariant bilinear form,

$$([x, y], z) = (z, [y, z]) \quad \forall x, y, z \in \mathfrak{g}$$

(Eg. the Killing form $\kappa(x, y) = \text{Tr}_{\mathfrak{g}} \text{ad}(x) \text{ad}(y)$ is invariant.)

Typically we normalise $(\cdot, \cdot)$ so that $(\alpha, \alpha) = 2$ for the long roots of $\mathfrak{g}$.

Then $\hat{\mathfrak{g}} = \mathfrak{g}[t, t^{-1}] \oplus \mathbb{C}K$ (affine Lie algebra),

$$[at^m, bt^n] = [a, b]t^{m+n} + m\delta_{m, -n}(a, b)K, \quad [K, \hat{\mathfrak{g}}] = 0$$

and $\tilde{\mathfrak{g}} = \hat{\mathfrak{g}} \oplus \mathbb{C}d$, $[d, K] = 0$, $[d, at^m] = mat^m$, (affine Kac-Moody algebra or “extended affine Lie algebra”)

Theorem 4.1. $\tilde{\mathfrak{g}}$ is a Kac-Moody algebra.

Let $\mathfrak{g} = \mathfrak{h} \oplus \bigoplus_{\alpha \in \Delta} \mathfrak{g}_{\alpha}$, ($\mathfrak{g}_{\alpha} = \mathbb{C}E_{\alpha}$.)

The simple roots and coroots. $\tilde{\mathfrak{h}} = \mathfrak{h} \oplus \mathbb{C}K \oplus \mathbb{C}d$. We identify $\tilde{\mathfrak{h}}^*$ with $\mathfrak{h}^* \oplus \mathbb{C}\Lambda_0 \oplus \mathbb{C}\delta$ where

$$\Lambda_0(\mathfrak{h}) = \delta(\mathfrak{h}) = 0$$

$$\Lambda_0(d) = \delta(K) = 0$$

$$\Lambda_0(K) = \delta(d) = 1$$

The *real coroots* are

$$\hat{\Delta}^{V, re} = \{E_{\alpha}t^m | \alpha \in \Delta, m \in \mathbb{Z}\}$$

and there are also imaginary roots and coroots

$$\hat{\Delta}^{V, im} = \{Ht^m | H \in \mathfrak{h}, m \in \mathbb{Z} \setminus \{0\}\}$$

Roots:

$$\hat{\Delta}^{re} = \{\alpha + m\delta | \alpha \in \Delta, m \in \mathbb{Z}\}$$

$$\hat{\Delta}^{im} = \{m\delta | m \neq 0\}$$

Xt^m :

$$[H, Xt^m] = [H, x]t^m, \quad H \in \mathfrak{h}$$

$$[K, xt^m] = 0$$

$$[d, xt^m] = mxt^m$$

so it $x \in \mathfrak{g}_{\alpha}$, $xt^m \in \tilde{\mathfrak{g}}_{\alpha+m\delta}$.

The invariant bilinear form $(\cdot, \cdot)$ from $\mathfrak{g} \times \mathfrak{g}$ extends uniquely to $(\cdot, \cdot) : \tilde{\mathfrak{g}} \times \tilde{\mathfrak{g}} \rightarrow \mathbb{C}$.

$(d, d) = (K, K) = 0$, $(d, K) = 1$ and $(d, \mathfrak{h}) = (K, \mathfrak{h}) = 0$.

So, in $\tilde{\mathfrak{h}}^*$:

$$\begin{aligned} (\Lambda_0, \Lambda_0) &= (\delta, \delta) = 0 \\ (\Lambda_0, \mathfrak{h}^*) &= (\delta, \mathfrak{h}^*) = 0 \\ (\Lambda_0, \delta) &= 1. \end{aligned}$$

Hence, $|\alpha + m\delta|^2 = |\alpha|^2$, $|m\delta|^2 = 0$.

Example 4.2. $\widetilde{\mathfrak{sl}}_2, \tilde{\mathfrak{h}}^* = \text{span}\{\alpha, \Lambda_0, \delta\}$ with Gram matrix ...

We can make a choice of positive roots,

$$\hat{\Delta}_+ = \{\alpha + m\delta | \alpha \in \Delta, m > 0\} \cup \{m\delta | m > 0\} \cup \Delta_+$$

Obviously, if $\alpha \in \Delta_+$ is simple, $\alpha \in \hat{\Delta}_+$ is simple.

Notation. Let $\theta \in \Delta_+$ be the highest root. ($\nexists \alpha \in \Delta_+$ such that $\alpha - \theta \in \mathbb{Z}_+ \Delta_+$.) and $\alpha = \delta - \theta$.

Then $\alpha_0 \in \hat{\Delta}_+$ is simple and the set of simple roots is $\hat{\Pi} = \{\alpha_0, \underbrace{\alpha_1, \dots, \alpha_\ell}_{\text{the finite simple roots}}\}$.

where $\ell = \text{rank}(\mathfrak{g})$.

The coroot corresponding to α_0 is

$$\alpha_0^\vee = K - \theta^\vee, \quad \theta^\vee = \frac{2}{(\theta, \theta)} \nu^{-1}(\theta) \in \mathfrak{h}$$

$$\text{and} \quad E_{\alpha_0} = E_{-\theta}t.$$

5. WEYL GROUP

Upshot. The Weyl group is a semidirect product of pseudoreflections and translations.

In any Kac-Moody algebra, we have

$$\begin{aligned} \text{roots} \quad \Pi &= \{\alpha_1, \dots, \alpha_\ell\} \subset \mathfrak{h}^* \\ \text{coroots} \quad \Pi^\vee &= \{\alpha_1^\vee, \dots, \alpha_\ell^\vee\} \subset \mathfrak{h}, \end{aligned}$$

and *reflections* $r_i \in \text{GL}(\mathfrak{h}^*)$, defined by

$$r_i(\lambda) = \lambda - \langle \lambda, \alpha_i^\vee \rangle \alpha_i.$$

One can check that

$$(r_i \lambda, r_i \mu) = (\lambda, \mu) \quad \forall \lambda, \mu \in \mathfrak{h}^*$$

The *Weyl group* W is $\langle r_i | i = 1, \dots, \ell \rangle \subset \text{GL}(\mathfrak{h}^*)$.

Example 5.1. For $\widetilde{\mathfrak{sl}}_2$, r_1 is easy,

$$\begin{aligned} r_1(\alpha) &= -\alpha \quad (\text{as in } \mathfrak{sl}_2) \\ r_1(\delta) &= \delta, \quad r_1(\Lambda_0) = \Lambda_0. \end{aligned}$$

To compute r_0 take an arbitrary element $m\alpha_1 + k\Lambda_0 + f\delta$ and do:

$$\begin{aligned} r_0(m\alpha_1 + k\Lambda_0 + f\delta) &= m\alpha_1 + k\Lambda_0 + f\delta - \langle \alpha_0^\vee, m\alpha_1 + k\Lambda_0 + f\delta \rangle \alpha_0 \\ \alpha_0 &= \delta - \alpha_1, \quad \alpha_0^\vee = K - \alpha^\vee \end{aligned}$$

so we obtain

$$\begin{aligned} &= m\alpha_1 + k\Lambda_0 + f\delta - (k - 2m)(\delta - \alpha_1) \\ &\quad (k - m)\alpha_1 + k\Lambda_0 + (f - k + 2m)\delta. \end{aligned}$$

Relative to basis $\{\alpha_1, \Lambda_0, \delta\}$.

$$\begin{aligned} r_1 &= \begin{pmatrix} -1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}, r_0 = \begin{pmatrix} -1 & 1 & 0 \\ 0 & 1 & 0 \\ 2 & -1 & 1 \end{pmatrix} \begin{pmatrix} m \\ k \\ f \end{pmatrix} \\ t &= r_1 r_0 = \begin{pmatrix} 1 & -1 & 0 \\ 0 & 1 & 0 \\ 2 & -1 & 1 \end{pmatrix} \end{aligned}$$

Notice that δ is fixed by all r_i . Also $m\alpha + k\Lambda_0 + f\delta$, the *coefficient* of Λ_0 is fixed by all r_i .

Then

$$t(m\alpha_1 + k\Lambda_0 + f\delta) = (m - k)\alpha_1 + k\Lambda_0 + (f - k + 2m)\delta.$$

Think of t as a translation.

The number k in

$$\mathfrak{h}^* \ni \hat{\lambda} = \lambda + k\Lambda_0 + f\delta$$

is called the *level* of $\hat{\lambda}$.

$\hat{\mathfrak{h}}$ = union of (hyper)planes of constant level which are stable under W . The roots α are all of level 0.

[Picture] “ r_1 changes the sign of the finite path”. And $t = r_1 r_0$ is a sort of translation. Indeed, in general we can consider $t_{\alpha_i} = r_{\alpha_i} \circ r_0 \in W$,

$$t_{\alpha}(\beta + m\delta) = \beta + (m + (\beta, \alpha_i))\delta$$

One can describe the action of t_{α} on $\hat{\lambda}$ in general (e.g. see [Kac90, Chapter 6])

Proposition 5.2. *For the affine Kac-Moody algebra $\hat{\mathfrak{g}}$ with $\hat{W} = \langle r_0, r_1, \dots, r_{\ell} \rangle$ its Weyl group (and $W = \langle r_1, \dots, r_{\ell} \rangle \subset \hat{W}$ the Weyl group of $\mathfrak{g}$), then $\hat{W} \simeq W \times t_{Q^{\vee}}$ (where it should be semidirect product instead of $\times \dots$) where $Q^{\vee}$ is the coroot lattice of $\mathfrak{g}$.*

Remark 5.3. For general Kac-Moody algebras, the Weyl groups are much larger, hyperbolic reflection groups.

In the affine case, $\hat{W}$ fixes level k , and $|\hat{\lambda}|$. One gets, in the intersection, paraboloids [Picture of section of hyperboloid that is a parabola].

6. WEYL CHARACTER FORMULA

Highest weight representations of Kac-Moody algebras. Let $\lambda \in \mathfrak{h}^*$, where $\mathfrak{g}(A) = \mathfrak{n}_- \oplus \mathfrak{h} \oplus \mathfrak{n}_+$ is a Kac-Moody algebra. We define a *Verma module*

$$M(\Lambda) = U(\mathfrak{g}) \otimes_{U(\mathfrak{h} + \mathfrak{n}_+)} \mathbb{C}v_{\Lambda}$$

where $\mathfrak{h} + \mathfrak{n}_+$ acts on V_{Λ} by:

$$\begin{aligned} Xv_{\Lambda} &= 0, & \forall x \in \mathfrak{n}_+, \\ Hv_{\Lambda} &= \Lambda(H)v_{\Lambda}, & \forall H \in \mathfrak{h} \end{aligned}$$

So $\mathbb{C}v_\Lambda$ is a $U(\mathfrak{h} + \mathfrak{n}_+)$ -module,

$$\begin{array}{c} U(\mathfrak{h} + \mathfrak{n}_+) \\ \downarrow \\ U(\mathfrak{g}) \end{array}$$

By the PBW theorem, $M(\Lambda)$ has a linear $\mathbb{C}$ -basis.

Let $\{F_{\alpha,i} : i = 1, \dots, \dim \mathfrak{g}_\alpha\}$ be a basis of $\mathfrak{g}_{-\alpha}$, $\forall \alpha \in \Delta_+$. Also choose a total order on Δ_+ . (Some sort of lexicographical order that takes longer to write than to say.)

$$F_{\alpha_1, i_1}, F_{\alpha_2, i_2}, \dots, F_{\alpha_s, i_s}, v_\Lambda$$

$$\alpha_1 \leq \alpha_2 \leq \dots \leq \alpha_s \text{ and if } \alpha_p = \alpha_{p+1}, i_p \leq i_{p+1}$$

We have $M(\Lambda)_\lambda = \{m | Hm = \lambda(H)m\}$ weight spaces.

$$M(\Lambda) = \bigoplus_{\lambda \in \mathfrak{h}^*} M(\Lambda)_\lambda$$

The vector v_Λ is in $M(\Lambda)_\Lambda$ by definition,

$$F_{\alpha,i} v_\Lambda \in M(\Lambda)_{\Lambda - \alpha}$$

$$H(Fv_\Lambda) = \underbrace{[H, F]v_\Lambda}_{= -\alpha(H)Fv_\Lambda} + \underbrace{FHv_\Lambda}_{=\Lambda(H)Fv_\Lambda}$$

So $\chi_{M(\Lambda)} = \sum_{\lambda \in \mathfrak{h}^*} \dim M(\Lambda)_\lambda e^\lambda$ is computed by counting monomials y with fixed $\sum_i \alpha_i$.

$$(6.0.1) \quad \chi_{M(\Lambda)} = e^\Lambda \prod_{\alpha \in \Delta_+} \frac{1}{(1 - e^{-\alpha})^{\dim \mathfrak{g}_\alpha}}.$$

The product on Eq. 6.0.1 is called *Weyl denominator*.

Exercise 6.1. Convince yourself of this.

Example 6.2. $\mathfrak{g} = \mathfrak{sl}_2$, [Picture]

$$\begin{aligned} \chi_{M(\Lambda)} &= e^\Lambda + e^{\Lambda - \alpha} + e^{\Lambda - 2\alpha} + \dots \\ &= e^\Lambda (1 + e^{-\alpha} + e^{-2\alpha} + \dots) \\ &= e^\Lambda \frac{1}{1 - e^{-\alpha}}. \end{aligned}$$

For certain Λ , $M(\Lambda)$ is *reducible* (i.e. there exists a submodule $0 \neq N \subset M(\Lambda)$ (with proper contention).

Lemma 6.3. For any submodule N ,

$$N = \bigoplus_{\mu \in \mathfrak{h}^*} N \cap M(\Lambda)_\mu.$$

Corollary. The sum of all proper submodules of $M(\Lambda)$ is proper, in particular there is a maximal proper submodule.

Notation. $L(\Lambda) = M(\Lambda) / \left(\begin{smallmatrix} \text{max. proper} \\ \text{submodule} \end{smallmatrix} \right)$

Example 6.4. $\mathfrak{sl}_2$. $\Lambda = 3\omega$ (ω : fundamental weight, $\alpha = 2\omega$.) $L(3\omega) = \mathbb{C}\langle e^{3\omega}, e^{-\omega}, e^{-3\omega} \rangle$.
[Picture]

Definition 6.5. A $\mathfrak{g}$ -module is *integrable* if

- $V = \bigoplus_{\mu \in \mathfrak{h}^*} V_\mu$ (weight module).
- For all simple roots α_i ; e_i and f_i are locally nilpotent on V (i.e. for all $v \in V$ there exists N such that $e_i^N v = f_i^N v = 0$.)

Remark 6.6. • Vermas are not integrable.

- $\dim V < \infty \implies V$ integrable.
- $\mathfrak{g}$ itself (Kac-Moody) is integrable.

Dominant integrable weights. Let $\{\alpha_1^\vee, \dots, \alpha_\ell^\vee\} \subset \mathfrak{h}$ be the simple coroots.

Definition 6.7. The *dominant integral weights* are the weights that pair with the coroots to give integers:

$$P_+ = \{\lambda \in \mathfrak{h}^* : \langle \lambda, \alpha_i^\vee \rangle \in \mathbb{Z}_{\geq 0}, i = 1, \dots, \ell\}.$$

For $L(\Lambda)$ to be integrable, it is necessary that $\Lambda \in P_+$.

Indeed, suppose $L(\Lambda)$ is integrable. Then $f_i^N v_\Lambda = 0$ in $L(\Lambda)$, or rather

$$\underbrace{e_i f_i^{N+1} v_\Lambda}_{K f_i^N = 0} \in M(\Lambda),$$

and K can only be zero if $\langle \Lambda, \alpha_i^\vee \rangle \in \mathbb{Z}_{\geq 0}$. Applying for all i , we find $\Lambda \in P_+$ is necessary.

Proposition 6.8. $L(\Lambda)$ is integrable if and only if $\Lambda \in P_+$.

Proof. For the converse, use induction and Serre relations (we know the result for the highest weight, and want to prove for others). $\square$

Example 6.9. $\mathfrak{sl}_3$. [Picture]

Example 6.10. $\widehat{\mathfrak{sl}_2}$. [Picture, P_+ looks like diagonal lines.]

$$\alpha_0^\vee = K - H, \quad \alpha_1^\vee = H \in \mathfrak{sl}_2, \quad \langle \delta, \alpha_i^\vee \rangle = 0, \quad i = 0, 1$$

Remark 6.11. For affine Kac-Moody algebras, almost nothing about the structure of $M(\Lambda)$ depends on the coefficient of δ in Λ . So it's common to consider

$$M(\Lambda) = M_k(\lambda) = M(k\Lambda_0 + \lambda), \quad \lambda \in \mathfrak{h}^*$$

where k , the level of Λ , is super important.

Then

$$\underbrace{\hat{P}_+}_{\substack{\delta\text{-coef.} \\ = 0}} = \bigcup_{k \in \mathbb{Z}_{\geq 0}} \{k\Lambda_0 + \lambda \mid \lambda \in P_+^k\} \{k\Lambda_0 + \lambda \mid \lambda \in P_+^k\}.$$

$$P_+^k = \{\lambda \in P_+ \mid \langle \lambda, \theta \rangle \leq k\} \subset P_+ \text{ for } \mathfrak{g}.$$

Consider $V = \bigoplus_{\mu \in \mathfrak{h}^*} V_\mu$ integrable, and $V_\lambda \neq 0$. Let $i \in \{1, \dots, \ell\}$. Consider $U = \bigoplus_{n \in \mathbb{Z}} V_{\lambda + n\alpha_i} \subset V$, and the action of e_i, f_i and $h_i = [e_i, f_i]$.

$\mathfrak{sl}_2 \curvearrowright U$, locally integrable. By structure of $\mathfrak{sl}_2$ -representations, U must be finite-dimensional with “symmetrical” weight space multiplicities, i.e.,

$$\{\lambda + n\alpha_i \mid n \in \mathbb{Z}\} \cap \{\text{weights of } V\} = \{\lambda + n\alpha_i \mid -p \leq n \leq q\}.$$

and

$$\langle \lambda - p\alpha_i, h_i \rangle = -\langle \lambda + q\alpha_i, h_i \rangle.$$

Consequently, the reflection $r_i(\lambda) := \lambda - \langle \lambda, \alpha_i^\vee \rangle \alpha_i$ has the same multiplicity as λ .

[Picture]

Now consider $M(\lambda)$ and $L(\Lambda)$ for $\Lambda \in P_+$. Actually, for general Λ , $M(\Lambda)$, while not necessarily irreducible, has an (Ω) -composition series* by irreducibles $L(\lambda)$.

$$M(\Lambda) = V_0 \supset V_1 \supset V_2 \supset \dots \supset V_n = 0,$$

such that V_i/V_{i+1} is $\simeq L(\lambda_i)$ (i.e. an irreducible highest weight module) for some $\lambda_i = \Lambda - \beta_i$, $\beta_i \in Q$. For Kac-Moody algebras we also consider the case that $(V_i/V_{i+1}) = 0$ for all $\mu \in \Omega + Q_+$, (which is the Ω -composition series).

[Picture.]

For an Ω -composition series, as above, we have

$$\text{ch}_{M(\Lambda)} = \sum_{\lambda \geq \Omega} \underbrace{[M(\Lambda) : L(\lambda)]}_{\substack{\# \text{ of times } L(\lambda) \\ \text{appears in the compos. series}}} \text{ch}_{L(\lambda)} \in \langle e^\mu : \mu \not\geq \Omega \rangle$$

Sending “ $\Omega \rightarrow -\infty$ ”, the identity

$$\text{ch}_{M(\Lambda)} = \sum_{\lambda \leq \Lambda} [M(\Lambda) : L(\lambda)] \text{ch}_{L(\lambda)}$$

makes sense.

Notation. $b_{\Lambda, \lambda} = [M(\Lambda) : L(\lambda)]$.

Remark 6.12. Recall the partial order on weights that $\lambda \leq \Lambda$ if $\Lambda - \lambda \in Q = \sum \mathbb{Z}_+ \alpha_i$. $b_{\Lambda, \lambda} = 1$ if $\lambda = \Lambda$ and $b_{\Lambda, \lambda} = 0$ if not ($\lambda \leq \Lambda$).

If we choose a total order on $\mathfrak{h}^*$, compatible with $\leq$. Then $\{b_{\Lambda, \lambda}\}$ is a lower triangular matrix with 1 on the diagonal. We can define $\{m_{\Lambda, \lambda}\}$ the *inverse matrix*. It's again lower triangular, 1 on the diagonal, and all $m_{\Lambda, \lambda}$ are integers (maybe negative now). And we have

$$(6.12.1) \quad \text{ch}_{L(\Lambda)} = \sum_{\lambda \leq \Lambda} m_{\Lambda, \lambda} \text{ch}_{M(\lambda)}.$$

Example 6.13. $\mathfrak{sl}_2$. $M(3) \underset{L(3)}{\supset} M(-5) \underset{L(-5)}{\supset} 0$. Since $M(-5)$ is already irreducible.

[Missing...]

We want to discover $m_{\Lambda, \lambda}$. In general massively difficult. For $\Lambda \in P_+$, $m_{\Lambda, \lambda}$ easy. Multiply Eq. 6.12.1 by R

$$R \text{ch}_{L(\Lambda)} = \sum_{\lambda \geq \Lambda} m_{\Lambda, \lambda} e^\lambda \cdot e^\rho R \text{ch}_{L(\Lambda)} = \sum_{\lambda \leq \Lambda} m_{\Lambda, \lambda} e^{\lambda + \rho}.$$

What's ρ ? It's $\rho \in \mathfrak{h}^*$ chosen so that $\langle \rho, \alpha_i^\vee \rangle = 1$, and it's called the *Weyl vector*.

Remark 6.14. For $\mathfrak{g}$ finite-dimensional, $\rho = \sum_{i=1}^\ell \omega_i$ necessarily. (And equals $\frac{1}{2} \sum_{\alpha \in \Delta_+} \alpha$.)

For $\mathfrak{g}$ finite dimensional and the affine $\hat{\mathfrak{g}}$, $\hat{\rho} = h^\vee \Lambda_0 + \rho$ works.

For $w \in W = \langle r_i | i = 1, \dots, \ell \rangle$, define $w(\text{ch}_V) = \sum \dim V_\mu e^{W(\mu)}$. We saw $w(\text{ch}_V) = \text{ch}_V$ if V integrable. In particular $w(\text{ch}_{L(\Lambda)}) = \text{ch}_{L(\Lambda)}$, $\Lambda \in P_+$.

Lemma 6.15. $m_{\Lambda, \lambda} = 0$ unless $\lambda + \rho = w(\Lambda + \rho)$ for some $w \in W$

Claim. $r_i(e^\rho R) = -e^\rho R$. So $w(e^\rho R) = \det(w)e^\rho R$ for all $w \in W$.

Proof.

$$R = \prod_{\alpha \in \Delta_+} (1 - e^{-\alpha})^{\text{mult}(\alpha)}.$$

Note that

- (1) $\text{mult}(r_i(\alpha)) = \text{mult}(\alpha)$ for all $\alpha \in \Delta$. (Since $\mathfrak{g}$ is integrable!)
- (2) $r_i(\Delta_+) = \{-\alpha_i\} \cup (\Delta_+ \setminus \{\alpha_i\})$.

Any $\alpha \in \Delta_+$ is of the form $\alpha = \sum_{i=1}^\ell k_i \alpha_i$. If $\alpha \neq \alpha_i$, some $k_{j_0} \neq 0$, $j_0 \neq i$ and

$$\begin{aligned} r_i(\alpha) &= \sum_j k_j \alpha_j \langle \alpha, \alpha_i^\vee \rangle \alpha_i \\ &= \sum_j k'_j \alpha_j \\ &= e^\rho (e^{-\alpha_i} - 1) \left(\prod_{\alpha \in \Delta_+ \setminus \alpha_i} \right) \\ &= -e^\rho R. \end{aligned}$$

for $k'_{j_0} = k_{j_0} > 0$. Can't have a mixture of signs, so $r_i(\alpha) \in \Delta_\perp$.
So

$$\begin{aligned} r_i(e^\rho R) &= \prod_{\alpha \in \Delta_+ \setminus \alpha_i} (1 - e^{-\alpha})^{\text{mult}(\alpha)} \cdot (1 - e^{t\alpha_i} \cdot e^{\rho - \alpha_i}) \\ r_i &= \rho - \langle \alpha_i^\vee, \rho \rangle \alpha_i = \rho - \alpha_i \end{aligned}$$

Hence

$$\sum_{\lambda \leq \Lambda} m_{\Lambda, \lambda} e^{\lambda + \rho}$$

is W -skew-invariant. (Lemma 6.15.)

Hence

$$\sum_{\lambda \leq \Lambda} m_{\Lambda, \lambda} e^{\lambda + \rho} = \sum_{w \in W} \det(w) \cdot e^{w(\Lambda + \rho)}$$

In conclusion, the *Weyl character formula*

$$\text{ch}_{L(\Lambda)} = \sum_{w \in W} \det(w) \cdot \frac{e^{w(\Lambda + \rho) - \rho}}{R}$$

□

Corollary (Weyl denominator formula).

$$e^\rho R = \sum_{w \in W} \det(w) e^{w(\rho)}.$$

Next time: Affine case, θ -functions. Modular forms (Poisson summation.)

7. CHARACTERS OF INTEGRABLE HIGHEST WEIGHT FOR AFFINE KAC-MOODY ALGEBRAS

Can be calculated using the Weyl character formula.

Recall: $\hat{\mathfrak{g}} = \mathfrak{g}[t, t^{-1}] \oplus \mathbb{C}K \oplus \mathbb{C}d$.

Dual Cartan $\hat{\mathfrak{h}}^* = \mathfrak{h}^* \oplus \mathbb{C}\Lambda_0 \oplus \mathbb{C}\delta$.

We consider weights $\Lambda = k\Lambda_0 + \lambda$, $\lambda \in \mathfrak{h}^*$, $k \in \mathbb{Z}_+$, $\lambda \in P_+^k$.

Simple roots: $\alpha_0 = \delta - \theta$, $\{\alpha_1, \dots, \alpha_\ell\} \subset \mathfrak{h}^*$.

The affine Weyl group is (the *semidirect* product!)

$$\widehat{W} = \langle r_0, \dots, r_\ell \rangle \cong W \times T$$

where $T = \{t_\alpha | \alpha \in Q\}$, where t_α is the translation

$$t_\alpha(\Lambda) = \Lambda + k\alpha - \left((\lambda, a) + k \frac{|\alpha|^2}{2} \right) \delta$$

$$\Lambda = k\Lambda_0 + \lambda.$$

Let's compute $\chi_{L(\Lambda_0)}$ using the Weyl character formula

$$\chi_{L(\Lambda)} = \frac{\sum_{w \in \widehat{W}} \varepsilon(w) e^{W(\Lambda + \rho) - \rho}}{R}$$

Firstly $R = \prod_{\alpha \in \hat{\Delta}} (1 - e^{-\alpha})^{\text{mult}(\alpha)}$

Recall

$$\begin{aligned} \hat{\Delta}_+ &= \{m\delta | m \in \mathbb{Z}_{\geq 1}\} \\ &\cup \{\alpha_1 + m\delta | m \in \mathbb{Z}_{\geq 0}\} \\ &\cup \{-\alpha_1 + m\delta | m \in \mathbb{Z}_{\geq 1}\} \end{aligned}$$

So let's write $e^{m\delta} = q^m$ (i.e. $q = e^{-\delta}$, is a symbol) and $y = e^{\alpha_1}$.

So

$$R = \prod_{n=1}^{\infty} \underbrace{(1 - y^{-1}q^{n-1})}_{\alpha + (n-1)\delta} \underbrace{(1 - q^n)}_{n\delta} \underbrace{(1 - yq^n)}_{-\alpha + n\delta}$$

Now let's express numerator also in terms of q and y

$$\hat{\rho} = h^\vee \Lambda_0 + \rho$$

$h^\vee$ dual Coxeter number for $\mathfrak{g}$. For $\mathfrak{g} = \mathfrak{sl}_2$, $h^\vee = 2$. (For $\mathfrak{g} = \mathfrak{sl}_n$, $h^\vee = n$ and $\mathfrak{g} = E_8$, $h^\vee = 30$.)

For $\mathfrak{sl}_2$, $\rho = \frac{1}{2}\alpha_1 = \omega_1$.

$\Lambda = \Lambda_0$, $\Lambda + \rho = 3\Lambda_0 + \omega_1$.

Using the formula, we find

$$\begin{aligned} t_{m\alpha_1}(3\Lambda_0 + \omega_1) - (\Lambda_0 + \omega_1) = \\ \dots, \Lambda_0 - 3\alpha, -2\delta, \Lambda_0, \Lambda_0 + 3\alpha_1 - 4\delta, \dots \end{aligned}$$

[Picture]

Notation. $w(\Lambda + \rho) := w \circ \Lambda$.

One finds

$$\sum_{w \in \hat{W}} \varepsilon(w) e^{w \circ \Lambda_0} = e^{\Lambda_0} \left(\underbrace{1 + y^3 q^4 + y^{-3} q^2 + \dots}_{+ \text{ signs because } w \in T} \underbrace{-y^{-1} - y^2 q^2 - \dots}_{- \text{ signs because } w \in T_\sigma} \right)$$

We find explicitly

$$\chi_{L(\Lambda_0)} = e^{\Lambda_0} \frac{\sum_{m \in \mathbb{Z}} y^{3m} q^{3m^2+m} - \sum_{m \in \mathbb{Z}} y^{3m-1} q^{3m^2-m}}{\prod_{n=1}^{\infty} (1 - y^{-1} q^{n-1})(1 - q^n)(1 - yq^n)}$$

Exercise 7.1. Put this formula in mathematica and confirm that [Picture]

$$\chi_{L(\Lambda_0)} = e^{\Lambda_0} (1 + q(y^{-1} + 1 + y) + q^2(y^{-1} + 2 + y) + (q^3(2y^{-1} + 3 + 2y) + \dots))$$

Appears that the central column here are *partitions*, i.e. $p(n) = \#$ of partitions of n . To see why recall the generating function $\sum_{n=0}^{\infty} q^n p(n) = \prod_{k=1}^{\infty} \frac{1}{1-q^k}$, and expand.

It would appear that

$$\chi_{L(\Lambda_0)} = \prod_{k=1}^{\infty} \frac{1}{1-q^k} \cdot \sum_{m \in \mathbb{Z}} y^m q^{m^2}$$

This identity is true, and we will see it has a vertex-algebra interpretation.

Remark 7.2. If we compute $L(0) = 1$ using the formula, we obtain

$$\prod_{n=1}^{\infty} (1 - yq^{n-1})(1 - q^n)(1 - y^{-1}q^n) = \sum (\text{exercise})$$

This identity is called the *Jacobi triple product identity*.

These functions

$$\sum_{n \in \mathbb{Z}} q^{n^2}, \sum_{n \in \mathbb{Z}} y^n q^{n^2}, \sum_{n \in \mathbb{Z}} y^{3n} q^{3n^2+n}, \text{ etc. } \dots$$

are all examples of θ -functions.

Remark 7.3. In the formula for $\chi_{L(\Lambda_0)}(y, q)$, we could put $y = 1$ to get

$$\begin{aligned} \chi_{L(\Lambda_0)}(q) &= 1 + 3q + 4q^2 + 7q^3 + \dots \\ &= \prod \frac{1}{1-q^k} \cdot \sum_{m \in \mathbb{Z}} q^{m^2} \end{aligned}$$

If one looks at $L(\Lambda_0 + \omega_1)$ (the other $\Lambda \in P_+^1$),

$$\chi_{L(\Lambda + \omega_1)} = \prod \frac{1}{1-q^k} \sum_{m \in \mathbb{Z}} q^{(m+1/2)^2-1/4}$$

8. θ -FUNCTIONS

Let's consider

$$\theta(\tau) = \sum_{n \in \mathbb{Z}} q^{n^2/2}, \quad q = e^{2\pi i \tau}$$

This converges (absolutely on compact regions in) the domain $\text{Im}(\tau) > 0$.

Consider the Fourier transform

$$\hat{g}(y) = \int_{-\infty}^{\infty} g(x) e^{2\pi i x y} dx$$

Theorem 8.1 (Poisson summation).

$$\sum_{n \in \mathbb{Z}} g(n) = \sum_{n \in \mathbb{Z}} \hat{g}(n)$$

Let's take $g(x, t) = e^{-\pi t x^2}$. Then $\theta(it) = \sum_{n \in \mathbb{Z}} g(n, t)$.

In this case

$$\hat{g}(y) = \sqrt{t} e^{-\pi y^2/t}$$

(integral of Gaussian).

So we conclude that

$$\theta\left(-\frac{1}{\tau}\right) = \sqrt{\frac{\tau}{i}} \theta(\tau)$$

Note also that

$$\theta(\tau + 2) = \theta(\tau)$$

because $q^{\frac{1}{2}} = e^{\pi i \tau}$.

So $\theta(\tau)$ is an example of a modular form.

What is a modular form? Let

$$\mathrm{SL}_2(\mathbb{Z}) = \left\{ \begin{pmatrix} a & b \\ c & d \end{pmatrix} \mid a, b, c, d \in \mathbb{Z}, \det = 1 \right\},$$

which acts on $\mathbb{H} = \{\tau \in \mathbb{C} \mid \mathrm{Im}(\tau) > 0\}$ by

$$\begin{pmatrix} a & b \\ c & d \end{pmatrix} \cdot \tau = \frac{a\tau + b}{c\tau + d}.$$

Definition 8.2. Let $\Gamma = \mathrm{SL}_2(\mathbb{Z})$ or some subgroup. A (weak) modular form (of weight k) is $f : \mathbb{H} \rightarrow \mathbb{C}$ holomorphic such that

$$(8.2.1) \quad f\left(\frac{a\tau + b}{c\tau + d}\right) = (c\tau + d)^k f(\tau).$$

If we demand that Γ contains $\begin{pmatrix} 1 & N \\ 0 & 1 \end{pmatrix}$ for some $N \geq 1$, (so $f(\tau + N) = f(\tau)$), and so $f(\tau) = \sum_{n=-\infty}^{\infty} a(n) q^{2\pi i n/2}$, for $f : \mathbb{H} \rightarrow \mathbb{C}$ satisfying Eq. 8.2.1 and such that $f(\tau)$ is “meromorphic at cusps”, i.e. $f(\tau) = \sum_{n \geq N_0} a(n) q^{n/N}$, etc.

Finally we would add a factor

$$f\left(\frac{a\tau + b}{c\tau + d}\right) = \varepsilon(A) (c\tau + d)^k f(\tau).$$

So in particular a weight-0 modular form is a modular function.

Denote $S = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}$. So $S\tau = -\frac{1}{\tau}$, and denote $T = \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}$, so $T\tau = \tau + 1$.

Then $\Gamma = \langle S, T^2 \rangle \subset \mathrm{SL}_2(\mathbb{Z})$.

We saw

$$\theta(T^2\tau) = \theta(\tau)$$

$$\theta(S\tau) = \sqrt{\frac{\tau}{i}} \theta(\tau) = i^{-1/2} (c\tau + d)^{1/2} \theta(\tau).$$

So $\theta(\tau)$ is a modular form of weight $1/2$, for the group $\Gamma = \langle S, T^2 \rangle$ with multiplier system $\varepsilon : \Gamma \rightarrow \mathbb{C}^\times$ defined by

$$\begin{aligned}\varepsilon(S) &= i^{-1/2} \\ \varepsilon(T^2) &= 1\end{aligned}$$

Theorem 8.3. *The Dedekind eta function*

$$\eta(\tau) = q^{1/24} \prod_{n=1}^{\infty} (1 - q^n), \quad q = e^{2\pi i \tau}$$

is also a modular form of weight $1/2$ for $SL_2(\mathbb{Z}) = \langle S, T \rangle$.

$$\begin{aligned}\eta\left(-\frac{1}{\tau}\right) &= \sqrt{\frac{\tau}{i}} \eta(\tau) \\ \eta(T\tau) &= e^{2\pi i/24} \eta(\tau).\end{aligned}$$

9. VERTEX ALGEBRAS

- (1) Recall $\hat{\mathfrak{a}} = \mathbb{C}K \oplus \bigoplus_{n \in \mathbb{Z}} \mathbb{C}h_n$ the *Heisenberg Lie algebra*, or *oscillator Lie algebra*,

$$[h_m, h_n] = m\delta_{m, -n}K, \quad [K, \hat{\mathfrak{a}}] = 0.$$

Remark 9.1. “It’s the simplest case of an affine Lie algebra: a 1-dimensional Lie algebra with a bilinear form”. It’s an example of the affine Lie algebra construction $\mathfrak{g} \rightsquigarrow \hat{\mathfrak{g}}$, where now $\mathfrak{a} = \mathbb{C}$, and

$$\begin{aligned}(\cdot, \cdot) : \mathfrak{a} \times \mathfrak{a} &\longrightarrow \mathbb{C} \\ (1, 1) &\longmapsto 1\end{aligned}$$

- (2) Witt Lie algebra

$$W = \bigoplus_{n \in \mathbb{Z}} \mathbb{C}D_n,$$

$$[D_m, D_n] = (m - n)D_{m+n}.$$

The oscillator algebra $\hat{\mathfrak{a}}$ has a representation which we have seen already:

$$H = \mathbb{C}[x_1, x_2, \dots]$$

$$h_n \mapsto \begin{cases} n \frac{\partial}{\partial x_n} & \text{if } n > 0 \\ x_{-n} & \text{if } n < 0, \\ 0 & \text{if } n = 0 \end{cases}, \quad k \mapsto \text{Id}$$

A picture of H [Picture].

Here I am introducing a grading of H :

$$H = \bigoplus_{n \in \mathbb{Z}_{\geq 0}} H_n, \quad H_n = \text{span}\{x_{m_1}, \dots, x_{m_s} \mid \sum m_s = n\}.$$

Remark 9.2. $\dim(H_n) = \# \{\text{integer partitions of } n\} = p(n)$ and $\sum_{n=0}^{\infty} \dim(H_n)q^n = \prod_{k=1}^{\infty} \frac{1}{1-q^k}$.

Remark 9.3 (Verma module style). H can be presented alternatively as

$$H = U(\hat{\mathfrak{a}}) \otimes_{U(\hat{\mathfrak{a}}_+)} \mathbb{C}1$$

where $\mathbb{C}1$ is a 1-dimensional representation of

$$\hat{\mathfrak{a}}_+ = \bigoplus_{n \geq 0} \mathbb{C}h_n \oplus \mathbb{C}K.$$

$$h_n \cdot 1 = 0 \quad \forall n \geq 0, \quad K \cdot 1 = 1$$

Exercise 9.4. H is irreducible.

Proof. We need to prove that there is no vector subspace $V \subset \mathbb{C}[x_1, x_2, \dots] = H$ such that $xV \subset V$ for all $x \in \hat{\mathfrak{a}}$. And then the proof is basically noticing that the orbit of the action of $\hat{\mathfrak{a}}$ on H is all of H . That is, if we had proper subspace of H , we can always find an element of $\hat{\mathfrak{a}}$ that takes some element in V to the one of the elements that is not in V . Indeed, by applying h_n for different values of positive n we can take any element of V to a constant. Then we apply h_n for different values of negative n to obtain any monomial. Then we add these monomials and obtain any polynomial in H . $\square$

Notation. Instead of 1 let's write $|0\rangle$.

Remark 9.5. We can easily generalise H to

$$H^\mu = U(\hat{\mathfrak{a}}) \otimes_{U(\hat{\mathfrak{a}}_+)} \mathbb{C}|\mu\rangle,$$

($\mu \in \mathbb{C}$), where $\hat{\mathfrak{a}}_+$ is the same, but now

$$h_n|\mu\rangle = \begin{cases} 0 & \text{if } n > 0 \\ \mu \cdot |\mu\rangle & \text{if } n = 0 \end{cases} \quad K|\mu\rangle = |\mu\rangle.$$

H^μ is again a (irreducible) $\hat{\mathfrak{a}}$ -module.

Generalise even more: Let $\mathfrak{h}$ be a finite-dimensional vector space, and $(\cdot, \cdot) : \mathfrak{h} \times \mathfrak{h} \rightarrow \mathbb{C}$ a symmetric bilinear form. $\hat{\mathfrak{h}} = \mathfrak{h}[t, t^{-1}] \oplus \mathbb{C}K$,

$$[at^m, bt^n] = m(a, b)\delta_{m, -n}K, \quad [K, \hat{\mathfrak{h}}] = 0.$$

Let $\mu \in \mathfrak{h}$. Define $H^\mu = U(\hat{\mathfrak{h}}) \otimes_{U(\hat{\mathfrak{h}}_+)} \mathbb{C}|\mu\rangle$, $\hat{\mathfrak{h}}_+ = \mathfrak{h}[t] \oplus \mathbb{C}K$, and

$$at^m \cdot |\mu\rangle = \begin{cases} (\mu, a)|\mu\rangle & \text{if } m = 0 \\ 0 & \text{if } m > 0. \end{cases}$$

Returning to $\hat{\mathfrak{a}} \curvearrowright H = H^\circ$. Introduce

$$L_m = \frac{1}{2} \sum_{k \in \mathbb{Z}} h_{m-k} h_k.$$

If $m \neq 0$, this sum is well-defined in $\text{End}(H)$.

Indeed, h_{m-k} and h_k commute ($m \neq 0$), and $\forall p(\underline{x}) \in H$, there exists N such that $p(\underline{x}) = p(x_1, \dots, x_{N-1})$, then $h_k \cdot p(x) = 0 \ \forall k \geq N$, and $h_{m-k} \cdot p(x) = 0 \ \forall m - k \geq N$.

So $(\sum h_{m-k} h_k) p(\underline{x})$ is a finite sum. Of course, the number of terms in the sum depends on $p(x)$ and can be arbitrarily large.

Exercise 9.6. If m, n and $m + n$ are not zero, then

$$[L_m, L_n] = (m - n)L_{m+n}$$

holds in $\text{End}(H)$.

(Trying to be a representation of W

$$D_n \mapsto L_n \in \text{End}(H).)$$

But $\sum_{k \in \mathbb{Z}} h_{-k} h_k$ is not well-defined. Indeed,

$$\begin{aligned} \left(\sum_{k \in \mathbb{Z}} h_{-k} h_k \right) 1 &= \sum_{k \geq 1} k \frac{\partial}{\partial x_k} (x_k 1) \\ &= (1 + 2 + 3 + 4 + \dots) 1 \end{aligned}$$

!

Normal ordering idea (“that physicist do”). Let’s cheat and redefine the product as

$$:h_{-k} h_k: = \begin{cases} h_{-k} h_k & \text{if } k \geq 0 \\ h_k h_{-k} & \text{if } k < 0. \end{cases}$$

Now $\sum_{k \in \mathbb{Z}} :h_{-k} h_k:$ is well-defined!

Notation. Let us consider series of the form

$$a(z) = \sum_{n \in \mathbb{Z}} a_{(n)} z^{-n-1}, \quad a_{(n)} \in \text{End}(V),$$

where V is a vector space.

Definition 9.7 (Most important in the course). We say $a(z)$ is a *quantum field* on V if, for every $v \in V$ there exists $N \in \mathbb{Z}$ such that $a_{(n)} v = 0 \ \forall n \geq N$.

Example 9.8. $V = H = \mathbb{C}[x_1, x_2, \dots]$,

$$a(z) = h(z) = \sum_{n \in \mathbb{Z}} h_n z^{-n-1}.$$

$$h(z) = \dots + 3z^{-4} \frac{\partial}{\partial x_3} + 2z^{-3} \frac{\partial}{\partial x_1} + x_1 + x_2 z + x_3 z^2 + \dots$$

Any fixed $v \in H$ is $v = p(x_1, \dots, x_{N-1})$ for some N . Then $h_N v = N \frac{\partial}{\partial x_N} v = 0$.

So $h(z) \curvearrowright H$ is a quantum field.

Definition 9.9. For a quantum field (or any series in fact) $a(z) = \sum_{n \in \mathbb{Z}} a_{(n)} z^{-n-1}$, define the *creation* and *annihilating operators*

$$a(z)_+ = \sum_{n \leq -1} a_{(n)} z^{-n-1}, \quad a(z)_- = \sum_{n \geq 0} a_{(n)} z^{-n-1}.$$

The quantum field condition solves the problem of the infinite series...

The normally order product of quantum fields $a(z) = \sum_{n \in \mathbb{Z}} a_{(n)} z^{-n-1}$, and $b(z) = \sum_{n \in \mathbb{Z}} b_{(n)} z^{-n-1}$ is.

$$:a(z)b(z): = a(z)_+ b(z) + b(z) a(z)_-.$$

Exercise 9.10 (Most important of the course). Show that if $a(z)$ and $b(z)$ are quantum fields, then $:a(z)b(z):$ is well-defined, and is a quantum field.

Next idea: in our example of $h(z) \curvearrowright H$, the coefficients h_n were coming from a Lie algebra, so we had relations $[h_m, h_n] = (\dots)$. (Indeed, $[h_m, h_n] = m\delta_{m,-n}\text{Id}_H$ in this case.) We would like to interpret such relations at the level of $h(z)$.

$$\begin{aligned} [h(z), h(z)] &= h(z)h(z) - h(z)h(z) \\ &= \sum_p \sum_{m+n=p} h_m h_n z^{-(m+n)-2} - \sum_{m+n=p} h_n h_m z^{-(m+n)-2} \\ &= \sum_p z^{-p-2} \left(\underbrace{\sum_{m+n=p} [h_m, h_n]}_{\text{infinite?}} \right) \end{aligned}$$

which is a bad idea, since that sum can be infinite. Better idea: change a variable:

$$[h(z), h(w)] = \sum_{m,n \in \mathbb{Z}} [h_m, h_n] z^{-m-1} w^{-n-1}.$$

In this example,

$$\begin{aligned} [h(z), h(w)] &= \sum_{m,n \in \mathbb{Z}} m\delta_{m,-n} z^{-m-1} w^{-n-1} I_H \\ &= \sum_{m \in \mathbb{Z}} z^{-m-1} w^{m-1} I_H. \end{aligned}$$

Observe (geometric series):

$$\begin{aligned} \frac{1}{z-w} &= \sum_{k \geq 0} z^{-k-1} w^k \quad (\text{convergent for } |z| > |w|) \\ \frac{1}{z-w} &= - \sum_{k \geq 0} w^{-k-1} z^k \\ &= - \sum_{k < 0} z^{-k-1} w^k \quad (\text{convergent for } |z| < |w|) \end{aligned}$$

So, in a sense

$$\sum_{k \in \mathbb{Z}} z^{-k-1} w^k \stackrel{“=”}{=} \frac{1}{z-w} - \frac{1}{z-w}.$$

This motivates us to introduce the expression

Definition 9.11. The *formal delta function* is

$$\delta(z, w) = \sum_{k \in \mathbb{Z}} z^{-k-1} w^k \in \mathbb{C}[[z, z^{-1}, w, w^{-1}]].$$

Why delta function? It behaves like Dirac delta. Recall that $\delta(x)$ is the *distribution* on $\mathbb{R}$ defined by $\delta[f(x)] = f(0)$ for all test functions $f(x)$. Every function $k(x)$ gives a distribution D_k .

$$D_k[f] = \int_{-\infty}^{\infty} k(x) f(x) dx,$$

$$f \in C_c^\infty(\mathbb{R}).$$

If δ were of the form D_k (it's not) it would have to look like [Picture, positive part of y axis, all x axis.

One can show that (it's a theorem by Plameli)

$$\delta(x) = \frac{1}{2\pi i} \lim_{\varepsilon \rightarrow 0^+} \left(\frac{1}{x - i\varepsilon} - \frac{1}{x + i\varepsilon} \right)$$

as distributions (i.e. limit taken in “distributional sense”).

(So that's why we called the delta function that way.)

Denote by $i_{z,w}$ the “expansion in positive powers of w ”. E.g.

$$i_{z,w} \frac{1}{z - w} = \sum_{k \geq 0} z^{-k-1} w^k,$$

similarly,

$$i_{z,w} \frac{1}{(z - w)^2} = \sum_{k \geq 0} k z^{-k-1} w^{k-1},$$

$$i_{z,w}(w^{-2}) = w^{-2}.$$

Exercise 9.12. So, in fact,

$$[h(z), h(w)] = i_{z,w} \frac{1}{(z - w)^2} - i_{w,z} \frac{1}{(z - w)^2} = \partial_w \delta(z, w).$$

What exactly is $i_{z,w}$?

Notation.

$\mathbb{C}[z]$	ring of polynomials
$\mathbb{C}[z, z^{-1}]$	ring of Laurent polynomials
$\mathbb{C}[[z]]$	ring of power series
$\mathbb{C}[[z, z^{-1}]]$	vector space (not ring!) of formal distributions
$\mathbb{C}((z))$	field of Laurent series

where the last is $\sum_{n \geq N} f a_n z^n$ for some N .

Since $\mathbb{C}((z))$ is a field, $\mathbb{C}((z))((w))$ is also a field.

There are natural inclusions

$$\mathbb{C}[z, w] \rightarrow \mathbb{C}((z))((w)) \rightarrow \mathbb{C}[[z^{\pm 1}, w^{\pm 1}]].$$

Let $\mathbb{C}(z, w)$ denote the fraction field of the domain $\mathbb{C}[z, w]$. By the property of $\mathbb{C}(z, w)$, there exists an embedding

$$\begin{array}{ccc} \mathbb{C}[z, w] & \xrightarrow{\quad} & \mathbb{C}((z))((w)) \\ & \searrow & \nearrow i_{z,w} \\ & \mathbb{C}(z, w) & \end{array}$$

The following diagram does not commute: [diagram]

We computed

$$[h(z), h(w)] = \partial_w \delta(z, w)$$

We can similarly compute

$$\begin{aligned} h(z)h(w) &= :h(z)h(w): + i_{z,w} \frac{1}{(z-w)^2} \\ h(w)h(z) &= :h(z)h(w): + i_{w,z} \frac{1}{(z-w)^2}. \end{aligned}$$

Notation. We write $\partial_w = \frac{\partial}{\partial w}$ and $\partial_w^{(j)} = \frac{1}{j!} \partial_w^j$.

Lemma 9.13. (1) *If we multiply the delta function with $(z-w)$ we get zero, that is,*

$$(z-w)\delta(z,w) = 0.$$

(2)

$$i_{z,w} \frac{1}{(z-w)^{j+1}} - i_{w,z} \frac{1}{(z-w)^{j+1}} = \partial_w^{(j)} \delta(z,w).$$

(3) $\forall j \geq 0$,

$$(z-w)^{j+1} \partial_w^{(j)} \delta(z,w) = 0.$$

(4) *Whenever $m \leq j$,*

$$(z-w)^m \partial_w^{(j)} \delta(z,w) = \partial_w^{(j-m)} \delta(z,w).$$

Proof.

□

Remark 9.14. All this is completely parallel to the Dirac δ -distribution $\delta(x)$.

$$x\delta(x) = 0, \quad x\delta'(x) = \delta(x), \text{ etc.}, \quad x^2\delta'(x) = 0.$$

10. THE RESIDUE PAIRING

Let $f(z) \in \mathbb{C}[z, z^{-1}]$, which is a vector space with basis $\{z^n : n \in \mathbb{Z}\}$.

An element of the dual vector space is a formal linear combination

$$\sum_{n \in \mathbb{Z}} c_n \varphi_n, \quad \text{where } \varphi_n(z^k) = \begin{cases} 0 & \text{if } k \neq n \\ 1 & \text{if } k = n \end{cases}$$

(no restriction on the $c_n \in \mathbb{C}$).

Identify $\sum c_n \varphi_n$ with

$$c(z) = \sum c_n z^{-n-1} \in \mathbb{C}[z^{\pm 1}]$$

so that φ acts on $f(z) = \sum f_n z^n \in \mathbb{C}[z^{\pm 1}]$, as

$$\varphi(f) = \sum_n c_n f_n = \text{Res}_z c(z) f(z) dz,$$

where

Definition 10.1. If U is a vector space and $a(z) = \sum_n a_n z^n \in U[[z^{\pm 1}]]$,

$$\text{Res}_z a(z) dz = a_{-1}$$

Let's record a few properties of $\text{Res}_z(\cdot) dz := \text{Res}_z(\cdot)$.

Lemma 10.2. (1) $\text{Res}_z f(z) \delta(z, w) = f(w)$.

(2) $\text{Res}_z f(z) (\partial_z g(z)) = -\text{Res}_z (\partial_z f(z)) g(z)$.

Proof. (1) Straightforward computation.

(2) Product rule, and uses that derivatives have no residues. □

So, in our example, $H, h(z)$ we see that

$$(z - w)^2[h(z), h(w)] = 0,$$

which is a nontrivial example: we really need to take the square, and also the bracket is not zero.

Definition 10.3. Two quantum fields $a(z)$ and $b(z)$ on a vector space V are *mutually local* if there exists N such that

$$(z - w)^N[a(z), b(w)] = 0.$$

Remark 10.4. See [?, Chapter 1] for physics motivation. More generally, if $D = \sum_{m,n} D_{m,n} z^m w^n \in U[[z^{\pm 1}, w^{\pm 1}]]$, we call D *local* if $(z - w)^N D = 0$ for some N .

Proposition 10.5. Let $a(z), b(z)$ be a local pair of quantum fields on V . Then

$$(10.5.1) \quad a(z)b(w) = :a(z)b(z): + \sum_{j=0}^{N-1} c_j(w) i_{z,w} \frac{1}{(z-w)^{j+1}}$$

and

$$(10.5.2) \quad b(w)a(z) = :a(z)b(z): + \sum_{j=0}^{N-1} c_j(w) i_{w,z} \frac{1}{(z-w)^{j+1}}$$

In particular,

$$(10.5.3) \quad [a(z), b(w)] = \sum_{j=0}^{N-1} c_j(w) \partial_w^{(j)} \delta(z, w).$$

Furthermore

$$(10.5.4) \quad c_j(w) = \text{Res}_z (z - w)^j [a(z), b(w)].$$

Proof. First we prove 10.5.3 using 10.5.4. □

Exercise 10.6. Deduce 10.5.1 and 10.5.2 from 10.5.3.

Notation. If $a(w)$ and $b(w)$ are quantum fields on V , we denote

$$a(w)_{(j)} b(w) = \text{Res}_z (z - w)^j [a(z), b(w)]$$

for $j \in \mathbb{Z}_{\geq 0}$.

Example 10.7. On H , $[h(z), h(w)] = \partial_w \delta(z, w) I_H$. So $\text{Res}_z \partial_w \delta(z, w) I_H = 0$ and $\text{Res}_z (z - w) \partial_w \delta(z, w) I_H = \text{Res}_z \delta(z, w) I_H = I_H$.

Note that I_H is a (very simple) quantum field.

The following definition generalizes the j -product to any integer, possible negative.

Definition 10.8. For $n \in \mathbb{Z}$, and $a(w), b(w)$ quantum fields, we define the n^{th} product $a(w)_{(n)} b(w)$ as

$$a(w)_{(n)} b(w) = \text{Res}_z [i_{z,w} (z - w)^n a(z) b(w) - i_{w,z} (z - w)^n b(w) a(z)].$$

Remark 10.9. (1) We recover the prior definition: if $n \geq 0$,

$$i_{z,w}(z-w)^n = i_{w,z}(z-w)^n = \sum_{r=0}^n \binom{n}{r} z^{n-r} (-w)^r,$$

a finite sum.

(2) (Exercise.) If $n = 1$, then

$$\begin{aligned} a(w)_{(-1)}b(w) &= a(w)_+b(w) + b(w)a(w)_- \\ &=: a(w)b(w): \end{aligned}$$

(3) (Exercise.) The more negative products are not something new: for $k \geq 0$,

$$a(w)_{(-k-1)}b(w) = (\partial_w^{(k)} a(w))b(w):$$

We have actually already used the following proposition:

Proposition 10.10. *If $a(w), b(w)$ are quantum fields, then $a(w)_{(n)}b(w)$ is also a quantum field for all $n \in \mathbb{Z}$.*

Definition 10.11. A *vertex algebra* consists of a vector space V , a set $\mathcal{F}$ of quantum fields on V , a nonzero vector $|0\rangle \in V$, and a linear map $T : V \rightarrow V$, such that

- (1) $T|0\rangle = 0$, and $[T, a(z)] = \partial_z a(z) \forall a(z) \in \mathcal{F}$.
- (2) V is spanned by $a_{(n_1)}^{i_1}, \dots, a_{(n_s)}^{i_s}|0\rangle$, where $a^{ij}(z) \in \mathcal{F}$ (and $s(z) = \sum a_{(n)} z^{-n-1}$ always).
- (3) All pairs $a(z), b(z) \in \mathcal{F}$ are mutually local. (We saw (ref?) that this is equivalent to $[a(z), b(w)] = \sum_{j=0}^{N-1} c_j(w) \partial_w^{(j)} \delta(z, w)$ and described the coefficients $c_j, \dots$)

We call $|0\rangle$ the *vacuum vector* and T the *translation operator*.

In fact, our Heisenberg example that we've been discussing so far is an example:

Example 10.12. $V = H = \mathbb{C}[x_1, x_2, \dots]$, $\mathcal{F} = \{h(z)\}$, $|0\rangle = 1$, $T = ?$, is a vertex algebra.

Answer 1. Recall $L(z) = \frac{1}{2} :h(z)h(z): = \frac{1}{2} h(z)_{(-1)}h(z)$, $L(z) = \sum_{m \in \mathbb{Z}} L_m z^{-m-z}$. In particular, $L_{-1} = \frac{1}{2} \sum_{k \in \mathbb{Z}} h_k h_{-1-k}$. Well, $T = L_{-1}$.

Answer 2. Given that T must satisfy $T1 = 0$, and $[T, h(z)] = \partial_z h(z)$, i.e.

$$[T, h_n] = -n h_{n-1},$$

and H is generated from 1 by $\{h_n | n \leq -1\}$, the action of T on H (if well-defined) is completely determined.

Exercise 10.13. Write a formula for $T(x_1^{m_1} x_2^{m_2} \dots x_s^{m_s})$.

11. A SECOND DEFINITION OF VERTEX ALGEBRA

Now we aim for a second definition of vertex algebras.

Let $(V, |0\rangle, T, \mathcal{F})$ be a vertex algebra. Define

$$\begin{aligned}\tilde{\mathcal{F}} &= \{\text{quantum fields } a(z) \text{ on } V \mid [T, a(z)] = \partial_z a(z)\} \\ \overline{\mathcal{F}} &= \bigcup_{k \geq 0} \mathcal{F}_k, \\ \mathcal{F}_0 &= \{I_V\}, \\ \mathcal{F}_k &= \{a(z)_{(n)}b(z) \mid a(z) \in \mathcal{F}, b(z) \in \mathcal{F}_{k-1}\} \\ \mathcal{F}' &= \{b(z) \in \tilde{\mathcal{F}} \mid a(z), b(z) \text{ is local}, \forall a(z) \in \mathcal{F}\}\end{aligned}$$

Where I_V is the identity of V considered as a quantum field. The idea is that this is like taking a subalgebra and taking the commutator over and over again.

Lemma 11.1. $a(z)_{(-1)}I_V = a(z)$.

Proof. Direct calculation. □

Lemma 11.2. $a(z)_{(-n-1)}I_V = \partial_z^{(n)}a(z)$.

Proof. Similar. □

Lemma 11.3 (Dong). *Suppose $a(z), b(z)$ and $c(z)$ are mutually local in pairs. Let $n \in \mathbb{Z}$. Then $a(z)_{(n)}b(z)$ and $c(z)$ is a local pair.*

Proof. Done in lecture. □

Therefore

$$\mathcal{F} \subset \overline{\mathcal{F}} \stackrel{\text{Dong}}{\subset} \mathcal{F}' \subset \tilde{\mathcal{F}}$$

Remark 11.4. To be sure, we should check $\overline{\mathcal{F}} \subset \tilde{\mathcal{F}}$.

$\mathcal{F}$ tilde is the translation invariant

Exercise 11.5. If $[T, a(z)] = \partial_z a(z)$, $[T, b(z)] = \partial_z b(z)$, then

$$[T, a(z)_{(n)}b(z)] = \partial_z(a(z)_{(n)}b(z)).$$

Theorem 11.6. (1) For any $a(z) \in \tilde{\mathcal{F}}$,

$$\begin{aligned}a(z)|0\rangle &= v + z \cdot Tv + \frac{z^2}{2}T^2v + \dots \\ &= e^{2T}v \quad (v \in V)\end{aligned}$$

Thus we define

$$\begin{aligned}s : \tilde{\mathcal{F}} &\longrightarrow V \\ a(z) &\longmapsto a(z)|0\rangle|_{z=0}\end{aligned}$$

(2)

$$s(z(z)_{(n)}b(z)) = a_{(n)}s(b(z)).$$

(3) Let $\mathcal{G} \subset \tilde{\mathcal{F}}$ be such that $s(\mathcal{G}) = V$, and suppose $a(z)$ is local with all $b(z) \in \mathcal{G}$. “If you commute with a large enough bunch of guys, you are close enough to being zero.” If $s(a(z)) = 0$, then $a(z) = 0$.

Proof. (1) We need to prove $a(z)|0\rangle$ does not have negative powers of z . So suppose it does, i.e. there is $n \geq 0$ such that $a_{(n)}|0\rangle z^{-n-1} \neq 0$.

Idea: use translation invariance to pair n and $-n$, and then the fact that $a(z)$ is a quantum field to get some vanishing.

So first we translate and find that:

$$Tv(z) = Ta((z)|0) = (Ta(z) - a(z)T)|0\rangle = \partial_z a(z)|0\rangle$$

so that

$$(11.6.1) \quad v_n = \frac{1}{n}Tv_{n-1}, \quad (n \neq 0).$$

So, $v_n \neq 0$ for some $n < 0 \implies v_{n-1}$ also not zero. But we also know by $a(z)$ being a quantum field that there exists N such that $a_{(n)}|0\rangle = 0$ for all $n \geq N$. So, we'd have a contradiction. So $v(z) = \sum_{n \geq 0} v_n z^n$.

Now Eq. 11.6.1 also implies

$$v_1 = Tv_0, v_0 = \frac{1}{2}Tv_1, \dots, v_n = \frac{1}{n!}T^n v_0.$$

So $a(z)|0\rangle = e^{zT}v_0$ ($v_0 \in V$).

(2) Consider

$$a(w)_{(n)}b(w)|0\rangle = \text{Res}_z(a(z)b(w)i_{z,w}(z-w)^n - \underbrace{b(w)a(z)i_{w,z}(z-w)^n}_{=0})$$

where that vanishing is because the $i_{w,z}$ expands the argument in positive powers, and $b(w)a(z)$ also has only positive powers since $a(z)|0\rangle$ also has only positive powers (and the residue picks the coefficient of the power -1). So we obtain

$$= \text{Res}_z a(z)b(w)i_{z,w}(z-w)^n.$$

But

$$\begin{aligned} s(a(w)_{(n)}b(w)) &= (\text{Res}_z s(z)b(w)i_{z,w}(z-w)^n|0\rangle)|_{w=0} \\ &= \text{Res}_z a(z)b(w)z^n|0\rangle|_{z=0} \\ &= \text{Res}_z z^n a(z)(s(b(w))) \\ &= \text{Res}_z z^n \sum_k a_{(k)}z^{-k-1}s(b) \\ &= a_{(n)}s(b). \end{aligned}$$

(3) By locality, there exists N such that

$$\begin{aligned} (z-w)^N a(z)b(w)|0\rangle &= (z-w)^N b(w)a(z)|0\rangle \\ &= (z-w)^N b(w)e^{2T}s(a) = 0 \end{aligned}$$

$$\begin{aligned} (z-w)^N a(z)b(w)|0\rangle|_{w=0} &= 0 \\ \implies z^N a(z)s(b) &= 0 \\ \implies a(z)v &= 0 \\ \implies a(z) &= 0 \end{aligned}$$

□

Corollary.

- (1) $S|_{\mathcal{F}'}$ is injective.
- (2) $S|_{\overline{\mathcal{F}}}$ is surjective.
- (3) $S|_{\mathcal{F}}$ is isomorphism $\overline{\mathcal{F}} \rightarrow V$.

So we define the inverse (linear) map $Y : V \rightarrow \overline{\mathcal{F}}$, i.e. for all $a \in V$, $Y(a, z)$ denotes $s^{-1}(a) \in \overline{\mathcal{F}}$.

- (4) $(V, |0\rangle, T, \overline{\mathcal{F}})$ is a vertex algebra, and

$$Y(a_{(n)}b, z) = Y(a, z)_{(n)}Y(b, z) \quad \forall a, b \in V.$$

So we have

$$\begin{array}{ccccccc} \mathcal{F} & \longrightarrow & \overline{\mathcal{F}} & \longrightarrow & \mathcal{F}' & \longrightarrow & \tilde{\mathcal{F}} \\ & & \downarrow & & \downarrow & & \downarrow s \\ & & V & \xrightarrow{\text{id}} & V & \xrightarrow{\text{id}} & V \end{array}$$

- Proof.* (1) Now we apply part 3 in the theorem to $a(z) \in \mathcal{F}'$, $\mathcal{G} = \overline{\mathcal{F}}$. If $s(a(z)) = 0$, then $a(z) = 0$. So $s : \mathcal{F}' \rightarrow V$ is injective (as in the diagram).
(2) By iterating the property in the theorem that $a_{(n)}s(b(z)) = s(a(z)_{(n)}b(z))$ so we can take out “one by one” in the following product to obtain

$$a_{(n_1)}^1 \dots a_{(n_s)}^s |0\rangle = s(a^1(z)_{(n_1)} \dots a^s(z)_{(n_s)}) I_V$$

This proves the double headed arrow in the diagram.

- (3) Now define $Y(-, z) : V \rightarrow \overline{\mathcal{F}}$.

$$s(Y(a, z)_{(n)}Y(b, z)) = a_{(n)}s(Y(b, z)) = a_{(n)}b.$$

- (4) Then all the axioms in our definition of vertex algebra are satisfied!

□

Remark 11.7. Let $(V, |0\rangle, T, \mathcal{F})$ be a vertex algebra. $\overline{\mathcal{F}}$ and Y as above. $\forall a, b \in V$,

$$(11.7.1) \quad [Y(a, z), Y(b, w)] = \sum_{j=0}^{N-1} c_j(w) \partial_w^{(j)} \delta(z, w).$$

$$c_j(w) = Y(a, w)_{(j)}Y(b, w) = Y(a_{(j)}b, w).$$

Which is like something we had written before. But now we know more. Let us extract the z^{-m-1}, w^{-n-1} coefficient of Eq. 11.7.1. We obtain

$$LHS = \sum_{m, n} [a_{(m)} z^{-m-1}, b_{(n)} w^{-n-1}] = [a_{(m)}, b_{(n)}]$$

$$RHS = \sum_j \left(\sum_k (a_{(j)}b)_{(k)} w^{-k-1} \right) \left(\sum_s \binom{s}{j} z^{-s-1} w^{s-j} \right)$$

The coefficient of $z^{-m-1} w^{-n-1}$ of RHS is, $s = m$,

$$\sum_j \binom{m}{j} w^{m-j} (a_{(j)}b)_{(k)} w^{-k-1} = \sum_j \binom{m}{j} (a_{(j)}b)_{(m+n-j)}$$

So we put this as a proposition:

Proposition 11.8 (Commutator formula). *In a vertex algebra $(V, |0\rangle, T, Y)$, we have*

$$[a_{(m)}, b_{(n)}] = \sum_{j \geq 0} \binom{m}{j} (a_{(j)}b)_{(m+n-j)} \quad \forall a, b \in V, \quad m, n \in \mathbb{Z}$$

Exercise 11.9. Prove that in a vertex algebra

(11.9.1)

$$[Y(a, z)Y(b, w)i_{z,w} - Y(b, w)Y(a, z)i_{w,z}](z - w)^n = \sum_{j \geq 0} Y(a_{(n+j)}b, w)\partial_w^{(j)}\delta(z, w).$$

Hint. Prove that the left hand side is local.

By extracting coefficients of Eq. 11.9.1, we obtain for all $a, b, c \in V$ and $m, n, k \in \mathbb{Z}$:

$$(11.9.2) \quad \begin{aligned} & \sum_{j \geq 0} \binom{m}{j} (a_{(n+j)}b)_{(m+k-j)}c \\ &= \sum_{j \geq 0} (-1)^j \binom{n}{j} (a_{(m+n-j)}b_{(k+j)})c - (-1)^n b_{(n+k-j)}a_{(n+j)}c \end{aligned}$$

This is called *Bocherd's identity*, and is the key ingredient in our second definition of vertex algebra (see [Kac01, Proposition 4.8(b)]):

Definition 11.10. A *vertex algebra* is a vector space V , a nonzero vector $|0\rangle \in V$, and a set of bilinear products $V \times V \rightarrow V$, $a, b \mapsto a_{(n)}b$, $n \in \mathbb{Z}$ such that

- (1) $\forall a, b \in V \exists N$ such that $a_{(n)}b = 0 \forall n \geq N$.
- (2) $\forall a \in V, |0\rangle_{(n)}a = \delta_{n,-1}a$.
- (3) $\forall a \in V, a_{(-1)}|0\rangle = a$ and $a_{(\geq 0)}|0\rangle = 0$.
- (4) Equation 11.9.2 holds $\forall a, b, c \in V, m, n, k \in \mathbb{Z}$.

12. CALCULATING VERTEX ALGEBRAS

We begin with some notation.

Definition 12.1. Let V be a vertex algebra and $a, b \in V$. We package the elements $a_{(0)}b, a_{(1)}b, \dots$ into a series

$$[a_\lambda b] = \sum_{j \geq 0} a_{(j)}b \frac{\lambda^j}{j!}$$

called the λ -*bracket* or *operator product expansion (OPE)* of a and b .

Recall that elements in V correspond to fields $Y(a, z)$ and $Y(b, z)$. They are local, and the coefficients of their bracket are $c_j(w) = Y(a_{(j)}b, w)$. So we put that information in the λ -bracket.

We also denote by $:ab:$ the vector corresponding to the normally ordered product $:Y(a, z)Y(b, z):$. Again, by last time's theorem, $:ab: = a_{(-1)}b$.

Example 12.2. $V = H = \mathbb{C}[x_1, x_2, \dots]$, $\mathcal{F} = \{h(z)\}$ where $h(z) = \sum_{n \in \mathbb{Z}} z^{-n-1}$ (so in this example we are denoting $h_n \equiv h_{(n)}$), where, as before,

$$h_n = \begin{cases} n \frac{\partial}{\partial x_n} & n > 0 \\ x_n & n < 0 \end{cases}$$

Recall we completed $\mathcal{F}$ to $\overline{\mathcal{F}}$ and discovered

$$\begin{array}{ccc} V & \xrightarrow{\simeq} & \overline{\mathcal{F}} \\ a(z)|0\rangle|_{z=0} & \xleftarrow{s} & a(z) \\ a & \xrightarrow{Y} & Y(a, z) \end{array}$$

For $I_V \in \overline{\mathcal{F}}$, where

$$I_V = \sum I_{V(n)} z^{-n-1}$$

$$I_{V(n)} = \begin{cases} I_V & \text{if } n = -1 \\ 0 & \text{if } n \neq -1 \end{cases}$$

we get $I_V|0\rangle|_{z=0} = |0\rangle$. So,

$$Y(|0\rangle, z) = I_V$$

This shows that the vacuum $|0\rangle$ corresponds to the identity.

Next $h(z)$ in our example. We have

$$\begin{aligned} h(z)|0\rangle &= \sum_{n \in \mathbb{Z}} h_n 1 z^{-n-1} \\ &= \sum_{k \geq 1} x^k z^{k-1} + \sum_{k \geq 1} k \frac{\partial}{\partial x_k} 1 \\ \implies h(z)|0\rangle|_{z=0} &= x_1 \\ \implies S(h(z)) &= x_1 \\ \implies Y(x, z) &= h(z). \end{aligned}$$

So, x_1 corresponds to the sort of “generating field” $h(z)$.

What about x_2 ? Well, $h(z) = h(z)_{(-1)} I_V$. There was a lemma (maybe Remark 10.9?) that

$$a(z)_{(-2)} I_V = \partial_z a(z).$$

So $\partial_z h(z) \in \overline{\mathcal{F}}$,

$$\begin{aligned} a(\partial_z h(z)) &= \partial \sum_{k \geq 1} x_k z^{k-1} \Big|_{z=0} \\ &= \sum_{k \geq 1} (k-1) x_k z^{k-2} \Big|_{z=0} \\ &= x_2. \end{aligned}$$

So

$$Y(x_{n+1}, z) = \partial_z^{(n)} h(z).$$

Continuing:

$$Y(x_3, z) = \frac{1}{2} \partial_z^2 h(z),$$

and in general

$$Y(x_{n+1}, z) = \partial_z^{(n)} h(z).$$

The next question is: what about x_1^2 ? The trick here is that x_1 is identified with h , so

$$\begin{aligned} Y(x_1^2, z) &= Y(x_1 \cdot x_1, z) \\ &= Y(h_{(-1)} x_1, z) \\ &= Y(x_1, z)_{(-1)} Y(x, z) \\ &=: h(z)h(z): . \end{aligned}$$

Recall that $L(z) = \frac{1}{2} :h(z)h(z):$. We have claimed (Exercise 9.6) that $[L_m, L_n] = (m-n)L_{m+n}$ when $m, n, m+n \neq 0$ where

$$\begin{aligned} L(z) &= \sum_{m \in \mathbb{Z}} L_m z^{-m-2} \\ &= \sum_{n \in \mathbb{Z}} L_{(n)} z^{-n-1}. \end{aligned}$$

What about the λ -bracket? Let's denote $h = x_1$, (so $Y(h, z) = h(z)$, $L = \frac{1}{2}x_1^2$ and $Y(L, z) = L(z)$).

Well, long ago we computed that

$$\begin{aligned} [h(z), h(w)] &= \partial_w \delta(z, w) I_V \\ &= \sum_{j \geq 0} Y(h_{(j)} h, w) \partial_w^{(j)} \delta(z, w). \end{aligned}$$

Then

$$\begin{aligned} Y(h_{(0)} h, w) &= 0 \\ Y(h_{(1)} h, w) &= I_V \\ Y(h_{\geq 2} h, w) &= 0 \\ \implies h_{(1)} h &= |0\rangle, \text{ and} \\ h_{(j)} h &= 0, \quad j = 0 \text{ and } j \geq 2. \end{aligned}$$

So we see that

$$[h_\lambda h] = \lambda |0\rangle.$$

We often omit “ $|0\rangle$ ” from these computations, as if it were “1”.

So

$$[h_\lambda h] = \lambda.$$

$[h_\lambda |0\rangle] = 0 = [|0\rangle_\lambda h]$ because $[h(z), I_V] = 0$. In fact $[x_\lambda |0\rangle] = [|0\rangle_\lambda x] = 0$ for all $x \in V$.

What about $h_{(0)} h$? Well, $h_{(0)} = 0$ by definition of $h(z)$. So $h_{(0)} h = 0$. Next, $h_{(j)} = j \frac{\partial}{\partial x_j}$ for $j > 0$, so

$$h_{(1)} h = \frac{\partial}{\partial x_1} (x_1) = 1 = |0\rangle.$$

So $[h_\lambda h] = \lambda$ again.

What I really want is $[L_\lambda L] = ?$ Let's use Bocherd's formula 11.9.2.

First let's compute $(h_{(-1)} h)_{(0)} L$, i.e. we are putting $n = -1$, $m = 0$ and $k = 0$.

[missing computations]

Exercise 12.3. Check $(h_{(-1)} h)_{(1)} h = 2h$ and $(h_{(-1)} h)_{\geq 2} h = 0$. Thus $[L_\lambda h] = Th + \lambda h$.

If we wanted, we could continue to calculate $[L_\lambda L]$ by putting $c = \frac{1}{2}x_1^2$ above instead of $c = x_1$.

We find

$$[L_\lambda L] = \underbrace{TL}_{L_{(0)}L} + \underbrace{2\lambda L}_{L_{(1)}L} + \underbrace{\frac{\lambda^3}{12}|0\rangle}_{L_{(3)}L}.$$

This means

$$(12.3.1) \quad [L(z), L(w)] = (\partial_w L(w))\delta(z, w) + 2L(w)\partial_w \delta(z, w) + \frac{1}{12}\partial_w^3 \delta(z, w)I_V.$$

So we can extract coefficients:

$$LHS = \sum_{m,n} [L_m, L_n] w^{-m-2} z^{-n-2}$$

Coefficient of $w^{-m-2} z^{-n-2}$ on RHS is:

$$\begin{aligned} & \sum_k (-k-2)L_k w^{-k-3} \sum_a z^{-a-1} w^a \\ & + 2 \sum_k L_k w^{-k-2} \sum_a a z^{-a-1} w^{a-1} + \frac{1}{12} \sum_a a(a-1)(a-2) z^{-a-1} w^{a-3}. \end{aligned}$$

So actually we found

$$[L_m, L_n] = (m-n)L_{m+n} + \frac{m^3-m}{12}\delta_{m,-n}I_V.$$

The moral is: one tries to define $L = \frac{1}{2}h^2$ naively, expecting $[L_m, L_n] = (m-n)L_{m+n}$. Need to “normally order/normalise” to make L well-defined, now $L = \frac{1}{2} :hh:$. The “cost” is that the expected relation $[L_m, L_n] = (m-n)L_{m+n}$ gets altered by an “anomaly” $\frac{m^3-m}{12}I_V$.

Right, so we showed how to compute this using Bochner’s formula, but there’s actually another way using λ -bracket:

Theorem 12.4. *Let V be a vertex algebra, $a, b, c \in V$. Then*

$$(12.4.1) \quad [Ta_\lambda b] = -\lambda[a_\lambda b]$$

$$(12.4.2) \quad [a_\lambda Tb] = (T + \lambda)[a_\lambda b]$$

$$(12.4.3) \quad T(:ab:) = :(Ta)b: + :a(Tb):$$

$$(12.4.4) \quad [b_\lambda a] = -[a_{-\lambda-T}b]$$

$$(12.4.5) \quad [a_\lambda :bc:] = :[a_\lambda bc:] + :b[a_\lambda c:] + \int_0^\lambda [[a_\lambda b]_\mu c] d\mu$$

$$(12.4.6) \quad [a_\lambda [b_\mu c]] - [b_\mu [a_\lambda c]] = [[a_\lambda b]_{\lambda+\mu} c].$$

Proof. For the first one use $\partial_z \delta(z, w) = -\partial_w \delta(z, w)$.

For the last one notice that

$$-[a_{-\lambda-T}b] = \sum_j \frac{1}{j!} (-\lambda - T)^j (a_{(j)}b)$$

$$\text{if } [a_\lambda b] = \sum_j \frac{\lambda^j}{j!} b.$$

□

Remark 12.5. By noting that $[a_\lambda b] = \text{Res}_z e^{\lambda z} Y(a, z)b$, these identities can be proved pretty efficiently.

That is,

Exercise 12.6. Use the fifth equation to prove what we proved with Bochner's formula.

And another one:

Exercise 12.7. Let $V = H = \mathbb{C}[x_1, x_2, \dots]$ again. Define $B = \frac{1}{2} :hh: + \beta Th$ with $\beta \in \mathbb{C}$. Confirm that

$$[B_\lambda B] = TB + 2\lambda B = \frac{1}{12}(1 - 12\beta^2)|0\rangle.$$

Definition 12.8. The *Virasoro Lie algebra* is

$$\text{Vir} = \bigoplus_{n \in \mathbb{Z}} \mathbb{C}L_n \oplus \mathbb{C}C$$

with Lie bracket

$$[L_m, L_n] = (m - n)L_{m+n} + \frac{m^3 - m}{12}\delta_{m, -n}C, \quad [C, \text{Vir}] = 0.$$

By the discussion above (all we've said so far!), there is a representation ρ of Vir in H given by

$$\rho(\underbrace{L_n}_{\substack{\text{as an abstract} \\ \text{object in Vir}}}) = \underbrace{L_n}_{\substack{\text{as a complicated} \\ \text{operator}}}, \quad \rho(C) = I_H.$$

Also H carries a representation ρ_β of Vir ,

$$\rho_\beta(L_n) = B_n, \quad \rho_\beta(C) = (1 - 12\beta^2)I_V$$

If (M, ρ) is a representation of Vir , in which $\rho(C) = c \cdot I_M$ for some scalar c , we say M has a *central charge* c .

Looks like we have a second example of a vertex algebra. Consider $V = H$, $|0\rangle = 1$, $T = T$ from before, but this time put $F = \{B(z)\}$. Then all the axioms of the first definition of vertex algebra are satisfied, except the second: $B_{(n_1)}B_{(n_2)} \dots B_{(n_s)}|0\rangle$ might not span all of $V = H$. So take $V = \text{span}\{\text{these monomials}\} \subset H$. One can check that $T(V) \subset V$, and $B(z)$ is a quantum field on V . So $(V, 1, T, \{B(z)\})$ is a vertex algebra, called the *Virasoro vertex algebra*.

13. THE CHARGED FREE FERMIONS: A VERTEX SUPERALGEBRA

Also known as:

- The Clifford (vertex) algebra.
- The Dirac sea.
- The bc system.

First we consider a vector superspace with basis φ and φ^*

$$\mathfrak{a} = \mathbb{C}\varphi + \mathbb{C}\varphi^*$$

(where φ and φ^* are odd, so this is a $\mathbb{O}, 2$ -dimensional superspace).

(A vector superspace is a $\mathbb{Z}/2$ -graded vector space, i.e. a vector space split in two pieces, $V = V_0 \oplus V_1$, which we call even and odd.)

We consider a bilinear form

$$\begin{aligned} \langle \cdot, \cdot \rangle : \mathfrak{a} \times \mathfrak{a} &\longrightarrow \mathbb{C} \\ \langle \varphi^*, \varphi \rangle &= 1 \\ \langle \varphi, \varphi^* \rangle &= 1 \text{ (why?)} \end{aligned}$$

Now consider

$$\tilde{\mathfrak{a}} = \underbrace{t^{1/2}\mathfrak{a}[t, t^{-1}]}_{\text{odd}} \oplus \underbrace{\mathbb{C}K}_{\text{even}}.$$

with Lie bracket

$$[at^m, b^n] = \delta_{m, -n} \langle a, b \rangle K, [K, \tilde{\mathfrak{a}}] = 0.$$

(Notice the small difference from the Heisenberg case: there is no m before the δ !)

To see why $\langle \varphi, \varphi^* \rangle = 1$, notice that in a Lie superalgebra we always want $[x, y] = (-1)(-1)^{p(x)p(y)}[y, x]$ where p denotes the parity. Since both at^m and bt^n are odd, we have $[at^m, bt^n] = [bt^n, at^m]$. Then apply the definition of bracket.

Since a, b odd here, we say $\langle a, b \rangle = \langle b, a \rangle$ means $\langle \cdot, \cdot \rangle$ is skew-super symmetric.

Definition 13.1. A bilinear form

$$(\cdot, \cdot) : U \times U \rightarrow \mathbb{C}$$

is *supersymmetric* if $(b, a) = (-1)^{p(a)p(b)}(a, b)$, and *skew-supersymmetric* if $(b, a) = -(-1)^{p(a)p(b)}(a, b)$.

Exercise 13.2. Invent the definition of Lie superalgebra, and confirm that if $U = U_0 \oplus U_1$ is a vector superspace, $\text{End}(U)$ is a Lie superalgebra with $[X, Y] := XY - (-1)^{p(X)p(Y)}YX$.

Let's build a Fock-type representation of $\tilde{\mathfrak{a}}$.

Construction 1.

$$\tilde{\mathfrak{a}}_+ = \bigoplus_{n \geq 0} t^n \mathfrak{a} \oplus \mathbb{C}K, \quad (n \in \frac{1}{2} + \mathbb{Z}, \text{ recall})$$

$\tilde{\mathfrak{a}}_+ \subset \tilde{\mathfrak{a}}$ is a superalgebra.

Consider $U(\tilde{\mathfrak{a}})$ and $U(\tilde{\mathfrak{a}}) \subset U(\tilde{\mathfrak{a}})$.

Recall that the PWB theorem says that $U(\mathfrak{g})$ “looks like” polynomials on $\mathfrak{g}$.

When $\mathfrak{g}$ is pure even, this says: take $\{a_1, a_1, \dots\}$ a basis of $\mathfrak{g}$, then

$$\{a_{i_1}^{m_1}, a_{i_2}^{m_2} \dots a_{i_s}^{m_s} : i_1 \leq i_2 \leq \dots \leq i_s, m_j \geq 1\}.$$

For $\mathfrak{g} = \mathfrak{g}_0 \oplus \mathfrak{g}_1$ super Lie algebra, the statement is similar, only now, we take each a^i homogeneous ($\in \mathfrak{g}_0$ or $\mathfrak{g}_1$):

$$\{a_{i_1}^{m_1} a_{i_2}^{m_2} \dots a_{i_s}^{m_s}, i_1 \leq i_2 \leq \dots \leq i_s, m_j \geq 1, \text{ if } a_{i_j} \text{ odd, } m_j = 1\}.$$

The point is that if we put an odd guy twice it becomes one.

Why? Because in $U(\mathfrak{g})$, $X \in \mathfrak{g}_1$ should satisfy $XX + XX = [X, X] = 0$. So $X^2 = 0$. So we only allow odd basis vectors to appear 0 or 1 times each.

Define $F = U(\tilde{\mathfrak{a}}) \otimes_{U(\tilde{\mathfrak{a}}_+)} \mathbb{C}|0\rangle$, where $at^n \cdot |0\rangle$ for all $n > \frac{1}{2}$, and $K|0\rangle = |0\rangle$.

[Picture of F]

We can introduce a bi-grading of F :

$$F = \bigoplus_{\substack{c \in \mathbb{Z} \\ \Delta}} \Delta$$

The point is that F is an infinite-dimensional vector space, but we get a reasonable mental picture of what it looks like.

What is $\dim F_{\Delta}^c$? Let's write...

So we have a vector superspace F , and an action on F of $\tilde{\mathfrak{a}}$. Let's build some quantum fields!

There are going to be two: $\varphi(z)$ and $\varphi^*(z)$.

$$\varphi(z) = \sum_{n \in \frac{1}{2} + \mathbb{Z}} \varphi_n z^{-n - \frac{1}{2}}$$

(Notice the exponent on z is an integer!)

$$\varphi^*(z) = \sum_{n \in \frac{1}{2} + \mathbb{Z}} \varphi_n^* z^{-n - \frac{1}{2}}.$$

Now let's compute $[\varphi^*(z), \varphi(w)]$.

$$\begin{aligned} [\varphi^*(z), \varphi(w)] &= \sum_{m, n \in \frac{1}{2} + \mathbb{Z}} [\varphi_m^*, \varphi_n] z^{-m - \frac{1}{2}} w^{-n - \frac{1}{2}} \\ &= \sum_{n \in \frac{1}{2} + \mathbb{Z}} K z^{n - \frac{1}{2}} w^{-n - \frac{1}{2}} = \sum_{\substack{r \in \mathbb{Z} \\ r = n - \frac{1}{2}}} z^r w^{-r - 1} = \delta(z, w). \end{aligned}$$

So $F, |0\rangle, \mathcal{F} = \{\varphi(z), \varphi^*(z)\}, T : F \rightarrow F$, is a vertex superalgebra. Here $T : F \rightarrow F$ has to satisfy $T|0\rangle = 0$ and $[T, \varphi(z)] = \partial_z \varphi(z)$,

$$\begin{aligned} \sum [T, \varphi_n] z^{-n - \frac{1}{2}} &= \sum \left(-n - \frac{1}{2}\right) \varphi_n z^{-n - \frac{3}{2}} \\ &= \sum \left(-k + \frac{1}{2}\right) \varphi_{k-1} z^{-k - \frac{1}{2}} \end{aligned}$$

after putting $n = k - 1$. So

$$[T, \varphi_n] = -\left(n - \frac{1}{2}\right) \varphi_{n-1}.$$

Inductively, this determines how T acts on F .

In λ -bracket notation, this just says

$$[\varphi_\lambda^* \varphi] = |0\rangle.$$

Skew-symmetry says $[b_\lambda a] = -(-1)^{p(a)p(b)}[a_{-\lambda-T} b]$.

For example,

$$\begin{aligned} [h_\lambda h] &= \lambda |0\rangle, \\ [h_\lambda h] &= -[h_{-\lambda-T} h] = -(-\lambda - T)|0\rangle \\ &= T\lambda |0\rangle. \end{aligned}$$

In this case $[\varphi_\lambda \varphi^*] = +|0\rangle$ too.

Notice $\sum \varphi_n z^{-n - \frac{1}{2}} = \sum \varphi_{(m)} z^{-m - 1}$.

So

$$\begin{aligned} \varphi_{(m)} &= \varphi_{m - \frac{1}{2}} \text{ and} \\ \varphi_{(m)}^* &= \varphi_{m - \frac{1}{2}}^*. \end{aligned}$$

So

$$\begin{aligned} :\varphi^* \varphi: &= \varphi_{(-1)}^* \varphi_{(-1)} |0\rangle \\ &= \varphi_{-\frac{1}{2}}^* \varphi_{-\frac{1}{2}} |0\rangle. \end{aligned}$$

I.e.

$$\varphi = \varphi_{(-1)}|0\rangle = \varphi_{-\frac{1}{2}}|0\rangle$$

Now we can put the former picture but with the associated operators using the second definition of vertex algebra: [Picture]

Let's compute some brackets: [lots of computations]

So,

$$[\alpha_\lambda \alpha] = \lambda|0\rangle.$$

Just like $[h_\lambda h] = \lambda|0\rangle$. This relation is the basis of the “boson-fermion correspondence”, i.e. consider the subspace $U \subset F$ defined as

$$U = \text{span}\{\alpha_{n_1}\alpha_{n_2}, \dots, \alpha_{n_s}|0\rangle : n_i \in \mathbb{Z}\}.$$

We claim that since $[T, \alpha(z)] = \partial_z \alpha(z)$, $[T, \alpha_n] = -n\alpha_{n-1}$.

So $T(U) \subset U$. Put $\mathcal{F}_1 = \{\alpha(z)\}$. Then $(U, |0\rangle, T|_U, \{\alpha(z)\})$ is a vertex algebra (an honest one, not super).

Proposition 13.3. *U is isomorphic to the Heisenberg vertex algebra. (In particular, $U \simeq \mathbb{C}[x_1, x_2, \dots]$ as a vector space.)*

Proof. Next time. □

Since $[\alpha_0, \varphi_n] = -\varphi_n$, $[\alpha_0, \varphi_n^*] = \varphi_n^*$, and $\alpha_0|0\rangle = 0$, in fact the charge of a monomial is exactly the eigenvalue of α_0 acting on it.

$$\begin{aligned} \alpha_0 \cdot \varphi_{-m}^* \varphi_{-m_2}^* \varphi_{-n}|0\rangle &= [\alpha_0, \varphi_{-m}^*, \varphi_{-m_2}^*, \varphi_{-n}]|0\rangle \\ &= \underbrace{(+1 + 1 - 1)}_{\text{charge}} \varphi_{-m_1}^* \varphi_{-m_2}^* \varphi_{-n}|0\rangle. \end{aligned}$$

From $[\alpha_\lambda \alpha] = \lambda|0\rangle$, we get

$$[\alpha_m, \alpha_n] = m\delta_{m, -n}I_F \text{ (as for } h\text{)}$$

So $[\alpha_0, \alpha_n] = 0$ for all $n \in \mathbb{Z}$. Thus $\alpha_0|_U = 0$ and thus $U \subset F^{(0)}$.

Proposition 13.4. $U = F^{(0)}$.

Proof. Next time. □

Let's take two copies of F , i.e. $F_2 = F \otimes F$ with fields $\varphi^1(z), \varphi^2(z), \varphi^{1*}(z), \varphi^{2*}(z)$

$$[\varphi_\lambda^{i*} \varphi^j] = \delta_{ij}.$$

Basis of F_2 is

$$\varphi_{-n_1}^1 \dots \varphi_{-n_2}^1 \varphi_{-m_1}^2 \dots \varphi_{-m_t}^2 \varphi_{-p_1}^{*-1} \dots \varphi_{-p_u}^{*-1} \varphi_{-q_1}^{*2} \dots \varphi_{-q_v}^{*2} |0\rangle.$$

Let $\alpha^{ij} = : \varphi^{j*} \varphi^i :$.

$$[\alpha_\lambda^{ij} \alpha^{k\ell}] = ?$$

(certainly $[\alpha_\lambda^{ii} \alpha^{ii}] = \lambda|0\rangle$.) In general

$$[\varphi_\lambda^k \alpha^{ij}] = \text{computation} = \delta_{ki} \varphi^j.$$

Similarly

$$[\varphi_\lambda^{k*} \alpha^{ij}] = -\delta_{kj} \varphi^{i*}.$$

$$[\alpha_\lambda^{ij} \varphi^k] = -\delta_{ik} \varphi^j$$

$$[\alpha_\lambda^{ij} \varphi^{k*}] = +\delta_{jk} \varphi^{i*}.$$

REFERENCES

- [Kac90] V.G. Kac, *Infinite-dimensional lie algebras*, Progress in mathematics, Cambridge University Press, 1990.
- [Kac01] ———, *Vertex algebras for beginners*, University lecture series, American Mathematical Society, 2001.